



Eigenvalues of The Robin Laplacian

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Declaration

This piece of work is a result of my own work and I have complied with the Department's guidance on multiple submission and on the use of AI tools. Material from the work of others not involved in the project has been acknowledged, quotations and paraphrases suitably indicated, and all uses of AI tools have been declared.

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Abstract

We explore the eigenvalues of the Laplacian subject to the Robin boundary condition. We examine specific instances in which the eigenfunctions can be explicitly found. We seek domains which extremise the first eigenvalue among certain classes of domain. We develop spectral theory in Hilbert spaces and apply it elliptic operators on Sobolev spaces. We then collect some properties of the eigenvalues and eigenfunctions, such as their monotonicity properties and nodal geometry. Lastly we apply everything to the more ambitious goal of finding extremisers of the first two eigenvalues among domains of a fixed volume.

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1 Prolegomenon

In Book I of the Aeneid, Virgil describes the flight of Dido from Phoenicia due to the murder of her husband by her brother, Pygmalion. She arrives in North Africa and supplicates a local leader to allow her as much land as “one ox-hide encircles”. With her request granted, she cuts an ox-hide into thin strips and uses it to establish the city of Carthage [33], as shown in Figure 1. While Virgil does not specify the exact shape of Carthage, Dido must surely have asked herself which shape of a fixed perimeter has the maximal area. While it was conjectured even in Dido’s era that the circle was the optimal shape, it was only in the 19th century that this was proven by Steiner using the method of symmetrisation. This result extends to higher dimensions and is known as the *isoperimetric inequality* [2]. Problems such as Dido’s inspire the field of shape optimisation and we will often ask the question ‘subject to some constraint, which shape extremises a certain functional?’. We will see cases where the extremisation is global, local, or even where no extremiser exists.



Dido Purches Land for the Foundation of Carthage. Engraving by Matthäus Merian the Elder, in Historische Chronick, Frankfurt a.M., 1630. Dido's people cut the hide of an ox into thin strips and try to enclose a maximal domain.

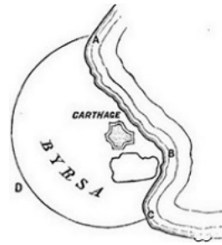


Figure 1: Dido founding Carthage (left) [2], a sketch of the optimal solution (right) [33].

1.1 The Wave Equation

We consider a connected domain (a bounded, open subset) in $\mathbb{R}$, namely an interval of the form (a, b) . A string spans this interval, lying flat at equilibrium, and is allowed to oscillate in a vertical plane. It is disturbed in some way, either by being struck (given some velocity), being deformed (given some displacement) or a combination of the two. There is a restoring force which acts to return the string to its initial configuration which is proportional to the curvature at each point [13]. Let us denote the height of the string relative to its equilibrium position by $\phi(x, t)$. Then the restoring force is equal to the local curvature of the string, multiplied by the tension τ in the string, which we assume to be constant. Newton’s second law tells us that this force is equal to the density ρ of the string, again assumed to be constant, multiplied by the acceleration. Equating these terms and defining $c^2 = \tau/\rho$ gives the one-dimensional wave equation:

$$\frac{\partial^2 \phi}{\partial t^2} = c^2 \frac{\partial^2 \phi}{\partial x^2}. \quad (1.1)$$

How do we interpret the value c^2 ? Dimensional analysis tells us:

$$\left[\frac{\partial^2 \phi}{\partial t^2} \right] = \frac{L}{t^2} = [c^2] \left[\frac{\partial^2 \phi}{\partial x^2} \right] = [c]^2 \frac{L}{L^2}.$$

Hence c has units $\frac{L}{t}$ so is a speed, the speed of propagation of our wave.

We’d like to extend this idea to regions in higher dimensions. We’ll begin by considering a membrane covering a domain $\Omega \subset \mathbb{R}^2$. The height of the membrane relative to its undisturbed

height is $\phi(x, y, t)$. To replicate the above procedure we would like a local measure of the curvature of the displacement height. It turns out that the Laplacian,

$$\nabla^2 \phi := \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2},$$

has all of the properties we desire. It is linear, invariant under translations and rotations of our coordinates, symmetric in x and y , and we recover the one-dimensional wave equation when $u_{yy} = 0$. This leads to the two-dimensional wave equation:

$$\frac{\partial^2 \phi}{\partial t^2} = c^2 \nabla^2 \phi. \quad (1.2)$$

We can extend the wave equation to any dimension by defining the N -dimensional Laplacian:

Definition 1.1. The Laplacian is the linear differential operator mapping $C^2(\Omega) \rightarrow C(\Omega)$ for some open, bounded domain $\Omega \subset \mathbb{R}^N$, expressed in Cartesian coordinates by

$$\nabla^2 = \sum_{i=1}^N \frac{\partial^2}{\partial x_i^2}.$$

The Laplacian can be rewritten in other coordinate systems as necessary.

We would like to solve equation (1.2) on a domain $\Omega \subset \mathbb{R}^N$ using the separation of variables. We begin by assuming that a solution takes the form $\phi(\mathbf{x}, t) = u(\mathbf{x})T(t)$. Substituting this into (1.2) gives:

$$T''(t)u(\mathbf{x}) = c^2 T(t) \nabla^2 u(\mathbf{x}).$$

Then dividing by $c^2 T(t)u(\mathbf{x})$ gives:

$$\frac{T''}{c^2 T} = \frac{\nabla^2 u}{u}.$$

Using the fact that the left-hand side is a function of t only and the right-hand side is a function of $\mathbf{x}$ only, both terms must be equal to a constant, say $-\lambda$. We can hence separate into a pair of equations:

$$T'' = -c^2 \lambda T, \quad (1.3)$$

$$-\nabla^2 u = \lambda u. \quad (1.4)$$

While equation (1.3) is easily solved given initial data, equation (1.4), known as the Helmholtz equation, is far more intricate and will occupy our attention herein. The solutions depend on both the domain Ω and the prescribed boundary conditions. We will explore the interplay between the possible values of λ and the domain and boundary conditions.

How do we interpret λ ? Dimensional analysis of the wave equation reveals that $\sqrt{\lambda}$ has units of frequency and so the values of λ correspond to the pure frequencies which can exist on a drumhead of shape Ω .

Looking at the method of separation of variables we can see that any partial differential equation of the form

$$\nabla^2 \phi(\mathbf{x}, t) = F(\phi_t, \phi_{tt}, \dots)$$

will separate into an ordinary differential equation for time and the spatial component will satisfy the Helmholtz equation. Other than the wave equation, another example of a well-studied equation which takes this form is the heat equation. If $\phi : \Omega \times [0, \infty) \rightarrow \mathbb{R}$ describes the heat distribution in a domain $\Omega \subset \mathbb{R}^N$ then ϕ obeys the heat equation:

$$\frac{\partial \phi}{\partial t} = k \nabla^2 \phi. \quad (1.5)$$

Here k is a parameter encoding the thermal diffusivity of the medium, which we will assume to be constant. Being able to interpret the solutions to equation (1.4) as the spatial parts of solutions to either the heat equation or the wave equation will turn out to be fruitful in providing physical intuition.

1.2 Boundary Conditions

For equation (1.4) to be well-posed on Ω (a bounded, open subset of $\mathbb{R}^N$), we must prescribe boundary conditions on the boundary of Ω , denoted $\partial\Omega$. Below $\frac{\partial u}{\partial \nu} = \nabla u \cdot \nu$ is the normal derivative to the boundary $\partial\Omega$ and ν is the outward pointing unit normal vector. For this to be defined almost everywhere we must consider domains whose boundaries are locally expressible as Lipschitz functions, referred to as Lipschitz domains. We introduce three common types of boundary condition:

1. Dirichlet: $u = 0$ on $\partial\Omega$,
2. Neumann: $\frac{\partial u}{\partial \nu} = 0$ on $\partial\Omega$,
3. Robin: $\frac{\partial u}{\partial \nu} + \alpha u = 0$ on $\partial\Omega$, $\alpha \in \mathbb{R}$.

Each condition has physical interpretations for both the wave and heat equations.

1. The Dirichlet condition is ‘fixed’: either the membrane is clamped along its boundary or it is held at a fixed temperature.
2. The Neumann condition on the other hand is ‘free’, allowing the membrane to vary at the boundary provided it remains ‘flat’ at each point. In the heat case the boundary is insulated: the temperature at the boundary can vary but no heat can flow in or out of it.
3. The Robin condition is a linear combination of the Dirichlet and Neumann ones and we see that we recover the Dirichlet condition in the limit $\alpha \rightarrow \infty$ (provided the normal derivative is bounded) and the Neumann condition in the limit $\alpha \rightarrow 0$. This tells us the Robin condition is a linear interpolation between the two extreme cases and so $\alpha > 0$ represents an elastically supported or partially insulated boundary.

The case $\alpha < 0$ is more difficult to interpret physically as α represents the spring constant of the elastic support. Examining the heat case sheds some light on this [38]. Newton’s law of cooling states that the heat flux from an object to its surroundings is proportional to the difference in temperatures between the two. Fourier’s law in one dimension states that the heat flux is proportional to the spatial derivative of the temperature. Thus at the right end point of a rod spanning $[0, l]$ held in a reservoir of constant temperature Q ,

$$\frac{\partial T}{\partial x}(l, t) = -\alpha(T(l, t) - Q),$$

for $\alpha > 0$. Intuitively, if $Q < T(l, t)$ Then heat flows out of the rod so the right-hand side should be positive. As $\alpha \rightarrow \infty$, the boundary radiates more of its heat to the surroundings, eventually losing all heat and recovering the Dirichlet condition. Rearranging gives

$$\frac{\partial T}{\partial x}(l, t) + \alpha T(l, t) = \alpha Q,$$

The Robin condition in one-dimension. Therefore we can interpret the negative α case as referring to an absorbing boundary condition, where heat is flowing into the boundary.

In full generality we can take α to be a function, however we will fix it to be a constant herein and the object of our study will be the *Robin problem*.

Definition 1.2. The Robin Problem on $\Omega \subset \mathbb{R}^N$ is the system

$$\begin{cases} -\nabla^2 u = \lambda u & \text{in } \Omega, \\ \frac{\partial u}{\partial \nu} + \alpha u = 0 & \text{on } \partial\Omega. \end{cases} \quad (1.6)$$

and the Robin Laplacian is the Laplacian operator with the domain of functions satisfying the Robin condition. Later we will define the domain more precisely.

1.3 Notation

Let us establish the notation which we will be using.

- All problems will be in $\mathbb{R}^N$, with N being the dimension of the ambient space.
- Specifically we will consider bounded, open subsets of $\mathbb{R}^N$, Ω , referred to as domains.
- Later we will see that the eigenvalues of the Robin problem form a countable set. Thus we can order them in increasing size and use $\lambda_n^\alpha(\Omega)$ to denote the n -th eigenvalue of the Robin Laplacian on the domain Ω with boundary parameter α , counted with multiplicity.
- In cases where we consider α to be fixed we might omit the dependence on α . Similarly we might omit Ω when the domain is fixed. When considering a general eigenvalue we might even drop the n .
- We will write $\mu_n(\Omega)$ and $\eta_n(\Omega)$ for the n -th Dirichlet and Neumann eigenvalues respectively, on the domain Ω .
- The normal vector field to the boundary $\partial\Omega$ of Ω is $\boldsymbol{\nu}$ and $\frac{\partial u}{\partial \boldsymbol{\nu}} := \nabla u \cdot \boldsymbol{\nu}$.
- We will denote by $B_r(x)$ the ball of radius r centred at x .
- The distance between a set $X \subset \mathbb{R}^N$ and a point $x_0 \in \mathbb{R}^N$ is $d(x_0, X) = \inf_{x \in X} |x - x_0|$, where $|\cdot|$ is the standard Euclidean distance.
- When evaluated on a set, $|\cdot|$ refers to both the N dimensional Lebesgue measure and the $N - 1$ dimensional Hausdorff measure. The surface integration measure is $d\sigma$.
- This shouldn't be confused with the notation we will use for a spectrum, $\sigma(T)$.
- Calligraphic letters will denote the domain, $\mathcal{D}(f)$, of a function f , the kernel $\mathcal{N} = \{x \in \mathcal{D}(f) : f(x) = 0\}$, and the range $\mathcal{R}(f) = \{f(x) : x \in \mathcal{D}(f)\}$.
- The support of a function, $\text{sppt}(f) = \{x \in \mathcal{D}(f) : f(x) \neq 0\}$, is the subset of its domain on which the function is non-vanishing.

Moving forward, we would first like to begin by exploring some specific scenarios in which the Laplacian eigenfunctions and eigenvalues can be explicitly calculated. In doing so we will collect some results on shape optimisation, specifically seeking domains extremising the first eigenvalue. After considering concrete examples which are highly symmetric, in Section 3 we broaden our view to the eigenvalues of general operators on Hilbert spaces. We then define the ‘correct’ Hilbert space to explore the eigenvalue problem, Sobolev spaces, and apply our Hilbert space theory. In Section 4 we use our results to see what information we can glean about the eigenvalues and eigenfunctions, exploring a variational characterisation, the monotonicity of the eigenvalues with respect to both the domain and the boundary parameter and the sets on which the eigenfunctions vanish. We conclude this section by collecting some results concerning the regularity and extrema of the eigenfunctions. With these results in hand we finally investigate the much broader question of the minimisation of the first two eigenvalues among domains of fixed volume and examine the proofs in depth.

In Chapter 4 of [15], which provides a survey of results pertaining to the Robin problem, the author writes

‘A single comprehensive, elementary treatment seems to be lacking in the standard literature.’

This report aims to fill that gap.

2 The Eigenvalue Problem on Symmetric Domains

The class of domains on which eigenfunctions can be explicitly calculated is remarkably small, each exhibiting a high degree of symmetry: it includes boxes, balls, tori and, lesser known, the equilateral triangle. Although I don't wish to provide exposition on the well-studied Dirichlet and Neumann problems, time and time again we will return to them for comparison to see the ways in which the Robin problem resists analogous analysis.

2.1 The Interval

The example of the Dirichlet and Neumann problems on arguably the simplest non-trivial domains, the interval and the box, are illustrative, adapted from the example given in [26]. The Dirichlet problem on the interval $I(t) = (-t, t)$ is given by

$$\begin{cases} -v''(x) = \mu v(x), & x \in I(t) \\ v(-t) = v(t) = 0. \end{cases}$$

It's not difficult to show that the eigenfunctions are given by

$$v_n(x) = \sin\left(\frac{n\pi}{2t}(x - t)\right),$$

with corresponding eigenvalue $\mu_n = \frac{n^2\pi^2}{4t^2}$ for $n \in \mathbb{N}$.

The Neumann problem is given by

$$\begin{cases} -w''(x) = \eta w(x), & x \in I(t) \\ w'(-t) = w'(t) = 0, \end{cases}$$

now with eigenfunctions

$$w_n = \cos\left(\frac{n\pi}{2t}(x - t)\right)$$

and corresponding eigenvalues $\eta_n = \frac{n^2\pi^2}{4t^2}$. It is important to note that now the index $n \in \mathbb{N} \cup \{0\}$ as the function $w_0 = 1$ is a permissible eigenfunction with eigenvalue 0, whereas $v_0 = 0$ is not an eigenfunction. We accordingly label the first Neumann eigenvalue by $\eta_0 = 0$. In both cases the eigenvalues form a sequence bounded from below and tending to infinity.

We now turn our attention to the Robin problem on the interval $I(t)$, following the work of Laugesen [25]:

$$\begin{cases} -u''(x) = \lambda^\alpha u(x), & x \in I(t) \\ -u'(-t) + \alpha u(-t) = u'(t) + \alpha u(t) = 0. \end{cases} \quad (2.1)$$

Note the difference in sign on the $u'(-t)$ due to the 'normal vector' being -1 at the left endpoint. If a function u is even or odd and satisfies the Robin boundary condition at the right end point then it automatically satisfies it at the left endpoint. For example if u is even, $u(-x) = u(x) \Rightarrow u'(-x) = -u'(x)$ and

$$u'(t) + \alpha u(t) = 0 \Rightarrow -u'(-t) + \alpha u(-t) = u'(t) + \alpha u(t) = 0.$$

Thus we can consider odd and even solutions to (2.1) separately. We also separately consider the cases when λ_1^α is positive, negative or zero.

$\lambda_1^\alpha > 0$: Write $\lambda_1^\alpha = \omega^2$. Then $u'' = -\omega^2 u$ has even solution $u = \cos(\omega x)$. The boundary condition at the right end point implies

$$-\omega \sin(\omega t) + \alpha \cos(\omega t) = 0 \Rightarrow \omega \tan(\omega t) = \alpha. \quad (2.2)$$

$\lambda_1^\alpha < 0$: Write $\lambda_1^\alpha = -\gamma^2$. Then $u'' = \gamma^2 u$ has even solution $u = \cosh(\gamma x)$. The boundary condition implies

$$\gamma \sinh(\gamma t) + \alpha \cosh(\gamma t) = 0 \Rightarrow \gamma \tanh(\gamma t) = -\alpha. \quad (2.3)$$

Notice that $\gamma \tanh(\gamma t) \geq 0$ so this has solutions for γ only when $\alpha < 0$. We will see this result in greater generality later. While equations (2.2) and (2.3) can't be solved analytically, their solutions are visualised in Figure 2.

$\lambda_1^\alpha = 0$: (2.1) has odd solution x and even solution 1. The even solution leads to $\alpha = 0$, reiterating the fact that the constant function is an eigenfunction of the Neumann Laplacian. The odd solution implies that $1 + \alpha t = 0$. We will not explore this case as we are interested in problems that involve fixing α and varying t .

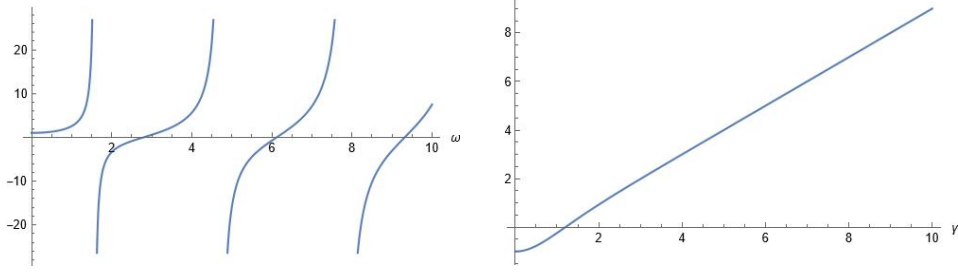


Figure 2: Solutions for ω and γ when $t = 1$ and $\alpha = -1$.

Remark 2.1. It is remarkable that there is a negative eigenvalue when $\alpha < 0$. For the interval (and in all cases as we will see later) both Dirichlet and Neumann eigenvalues are always non-negative. How do we interpret this negative eigenvalue? Returning to the heat case for a physical interpretation when $\alpha < 0$, we recall that the general solution to the heat equation for the temperature along a rod $I(t)$ is

$$T(x, t) = \sum_{n=1}^{\infty} A_n e^{-\lambda_n c t} u_n(x),$$

where (u_n, λ_n) are eigenpairs of the Laplacian [38]. When all eigenvalues are positive we see that the temperature decays exponentially in time, as physically intuitive. But if $\alpha < 0$ then our first eigenvalue is negative so the first term is exponentially increasing, meaning the temperature is unbounded everywhere. This agrees with the understanding that a negative boundary parameter embodies an absorbing, rather than radiating, boundary.

We can write our solutions to (2.2) and (2.3) more elegantly using the following definition:

Definition 2.1. Define the functions

$$\begin{aligned} g_1 &: [0, \pi/2) \rightarrow [0, \infty) \\ &: x \mapsto x \tan(x), \end{aligned}$$

$$\begin{aligned} h_1 &: [0, \infty) \rightarrow [0, \infty) \\ &: x \mapsto x \tanh(x). \end{aligned}$$

$$\begin{aligned} G_1 &: (0, \infty) \rightarrow (0, \pi/2) \\ &: y \mapsto \frac{g_1^{-1}(y)}{y}, \end{aligned}$$

$$\begin{aligned} H_1 &: (0, \infty) \rightarrow (0, \infty) \\ &: y \mapsto \frac{h_1^{-1}(y)}{y}. \end{aligned}$$

The utility of this definition is the following expression for the eigenvalues.

Lemma 2.1. *The first Robin eigenvalue on the interval $I(t)$ with boundary parameter α is given by*

$$\lambda_1^\alpha(I(t)) = \begin{cases} \alpha^2 G_1(\alpha t)^2 & \alpha > 0, \\ 0 & \alpha = -1/t, \\ -\alpha^2 H_1(-\alpha t)^2 & \alpha < 0. \end{cases}$$

Proof. This is merely a rewriting of the results (2.2) and (2.3). If $\alpha < 0$ then $\lambda_1^\alpha = -\gamma^2$ and (2.3) has a solution so this must be the minimal eigenvalue so we can write:

$$\gamma t \tanh(\gamma t) = -\alpha t \Rightarrow h_1(\gamma t) = -\alpha t \Rightarrow \gamma = \alpha \frac{h_1^{-1}(-\alpha t)}{\alpha t} = \alpha H_1(-\alpha t).$$

Similarly, in the positive α case $\lambda_1^\alpha = \omega^2$ so we rewrite (2.2):

$$\omega t \tan(\omega t) = \alpha t = g_1(\omega t) \Rightarrow g_1^{-1}(\alpha t) = \omega t \Rightarrow \omega = \alpha \frac{g_1^{-1}(\alpha t)}{\alpha t} = \alpha G_1(\alpha t).$$

We can find eigenvalues corresponding to the odd eigenfunctions but after establishing the necessary inequalities (*Lemma 5.1* and *Lemma 5.2* of [25]), we see that the ‘even eigenvalues’ are the minimal ones. $\square$

We now establish some properties of the functions G_1 and H_1 to shed light on how the first eigenvalue depends on the domain half length and boundary parameter. The functions are plotted in Figure 3.

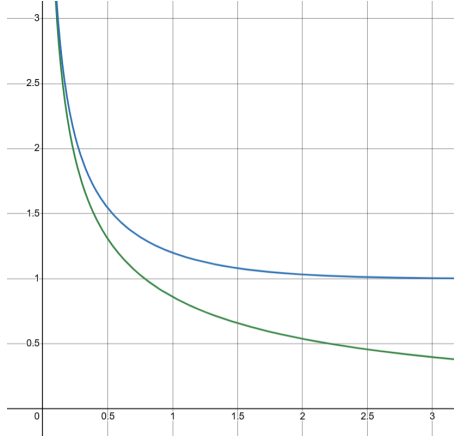


Figure 3: The functions $G_1(x)$ (green) and $H_1(x)$ (blue).

Lemma 2.2. *The functions g_1 and h_1 and their inverses are increasing.*

Proof. On the appropriate intervals $\tan$ and $\tanh$ are increasing and the products of increasing functions are also increasing. An invertible, increasing function has an increasing inverse by the Inverse Function Theorem. $\square$

Lemma 2.3. *The first eigenvalue is a strictly increasing function of α .*

We will later see that this holds in general.

Proof. This follows from the monotonicity results in Lemma 2.2.

$$\lambda_1^\alpha = \alpha^2 G_1(\alpha t)^2 = g_1^{-1}(\alpha t)^2 / t^2$$

is increasing in α . Also h_1^{-1} is increasing so $h_1^{-1}(-\alpha t)$ is positive and decreasing in α so $h_1^{-1}(-\alpha t)^2$ is too and lastly

$$-h_1^{-1}(-\alpha t)^2 / t^2 = -\alpha^2 H_1(-\alpha t)^2$$

is increasing in α . $\square$

Lemma 2.4. *The first eigenvalue is a strictly decreasing function of the interval half length, t .*

If $t_1 \leq t_2$ then $I(t_1) \subset I(t_2)$ and this lemma tells us that $\lambda_1(I(t_2)) \leq \lambda_1(I(t_1))$. This can be interpreted as a domain monotonicity result in one dimension, which we will see doesn't hold in higher dimensions.

Proof. In order to establish the increasing/decreasing nature of G_1 and H_1 , we use the fact that if $y = f(x)$ and $F(y) = \frac{f^{-1}(y)}{y}$ then

$$\frac{d}{dy}F(y) = \frac{d}{dy} \frac{f^{-1}(y)}{y} = \frac{dx}{dy} \frac{d}{dx} \frac{x}{f(x)} = \frac{1}{f'(x)} \left(\frac{x}{f(x)} \right)'.$$

The primes in the final expression represent derivatives with respect to x . Now turning to our functions G_1 and H_1 :

$$\left(\frac{x}{g_1(x)} \right)' = \cot(x)' = -\csc(x)^2, \quad (2.4)$$

$$\left(\frac{x}{h_1(x)} \right)' = \coth(x)' = -\operatorname{csch}(x)^2. \quad (2.5)$$

Thus both G_1 and H_1 are decreasing maps, from which the map $t \mapsto \lambda_1^\alpha(I(t))$ inherits decreasing monotonicity for α both positive and negative. $\square$

The functions G_1 and H_1 are decreasing, but what about their asymptotic behaviour around zero? Loosely speaking, as $t \rightarrow 0^+$, $\tan(t) = t + O(t^2) \Rightarrow t \tan(t) = t^2 + O(t^3)$ so the inverse function $g_1^{-1}(t) = \sqrt{t} + O(t)$ and $G_1(t) = t^{-1/2} + O(1)$ which is unbounded as $t \rightarrow 0^+$. The same asymptotic expansion holds for H_1 . Therefore Equation (23) in [25] states that

Lemma 2.5.

$$\lim_{t \rightarrow 0^+} G_1(t) = \lim_{t \rightarrow 0^+} H_1(t) = \infty.$$

A crucial idea we will need later to prove that something is a global minimum is that of *convexity*.

Definition 2.2. A set $X \subset \mathbb{R}^N$ is convex if for all $x, y \in X$ and $\tau \in [0, 1]$,

$$\tau x + (1 - \tau)y \in X.$$

Intuitively this means that for any two points in our set, the entirety of the line segment joining them must lie in the set too. For example, a disc is convex while an annulus is not.

Remark 2.2. Vector spaces are convex due to their closure under scalar multiplication and addition.

Definition 2.3. A function $f : X \subset \mathbb{R}^N \rightarrow \mathbb{R}$ is strictly convex on X if for all $x, y \in X$ and all $\tau \in (0, 1)$,

$$f(\tau x + (1 - \tau)y) < \tau f(x) + (1 - \tau)f(y).$$

Remark 2.3. The restriction of a strictly convex function to a convex subset $K \subset X$ is strictly convex. We require the subset to be convex as, denoting the restriction $f|_K$ and letting $x, y \in K$, $f|_K(\tau x + (1 - \tau)y)$ is only defined when $\tau x + (1 - \tau)y \in K$ i.e., when K is convex.

It is well known that a twice differentiable function mapping $\mathbb{R} \rightarrow \mathbb{R}$ with strictly positive second derivative is strictly convex.

Lemma 2.6. The functions $H_1(y)$ and $G_1(y)$ are convex with respect to $\log(y)$. That is,

$$\frac{d^2}{dz^2} G_1(e^z) > 0, \quad \frac{d^2}{dz^2} H_1(e^z) > 0.$$

Proof. The proof is simply a calculation of derivatives followed by establishing various trigonometric inequalities, the details of which are included in Lemma IV.8 of [25]. $\square$

2.2 The Box

Definition 2.4. The box in $\mathbb{R}^N$ is the Cartesian product of $N \in \mathbb{N}$ intervals,

$$B(\underline{t}) := I(t_1) \times \cdots \times I(t_N) \text{ for } \underline{t} \in \mathbb{R}^N.$$

A cube is a box with all edge lengths being equal, in any dimension.

We'll explore the Dirichlet and Neumann problem on the rectangle $I(t_1) \times I(t_2)$ but the idea generalises easily to arbitrary dimensions. A separation of variables argument with $u(x, y) = X(x)Y(y)$ reduces the problem to two copies of the Dirichlet/Neumann problem on each of the intervals $I(t_1)$ and $I(t_2)$ and so our eigenvalues are given by

$$\eta_{m,n} = \mu_{m,n} = \frac{m^2\pi^2}{4t_1^2} + \frac{n^2\pi^2}{4t_2^2}.$$

Again, the index set for $\mu_{m,n}$ is $(m, n) \in \mathbb{N} \times \mathbb{N}$ and as such the first Dirichlet eigenvalue is

$$\frac{\pi^2}{4t_1^2} + \frac{\pi^2}{4t_2^2},$$

whereas the index set for $\eta_{m,n}$ is $m, n \in \mathbb{N} \cup \{0\}$ so the first (non-trivial) Neumann eigenvalue is

$$\min \left\{ \frac{\pi^2}{4t_1^2}, \frac{\pi^2}{4t_2^2} \right\}.$$

Let's first examine the principal Dirichlet eigenvalue. Among rectangles of a fixed area (WLOG $t_1 t_2 = 1$), the principal eigenvalue can be made arbitrarily large by taking one of the edge lengths to be sufficiently small. On the other hand the eigenvalue is minimised when, setting $t_2 = \frac{1}{t_1}$,

$$\frac{d}{dt_1} \left(\frac{\pi^2}{4t_1^2} + \frac{\pi^2}{4} t_1^2 \right) = -2 \frac{\pi^2}{4t_1^3} + 2 \frac{\pi^2}{2} t_1 = 0 \Rightarrow t_1^4 = 1 \Rightarrow t_1 = t_2 = 1.$$

A second derivative test verifies that this is a minimum. Therefore the first Dirichlet eigenvalue is minimised by the square among rectangles of a fixed area. What about the principal Neumann eigenvalue? The area constraint is $t_1 t_2 = 1$ and without loss of generality we can take $t_1 \geq 1$. However,

$$\min \left\{ \frac{\pi^2}{4t_1^2}, \frac{\pi^2}{4} t_1^2 \right\}$$

does not exist for finite t_1 as we can make the first term arbitrarily small by taking t_1 to be suitably large. The more interesting question concerns maximising the principal Neumann eigenvalue. But the minimum of two objects is maximised when the objects are equal, that is when

$$\frac{\pi^2}{4t_1^2} = \frac{\pi^2}{4} t_1^2 \Rightarrow t_1 = t_2 = 1.$$

From now on our exploration will chiefly concern the Robin problem, with the Dirichlet and Neumann problems providing idealised comparisons.

Remark 2.4. Due to the fact that we can write an eigenfunction on the box $B(\underline{t})$ as a product of eigenfunctions on each interval $I(t_i)$,

$$\lambda_1^\alpha(B(\underline{t})) = \sum_{i=1}^N \lambda_1^\alpha(I(t_i)).$$

Lemma 2.7. *The first eigenvalue on the box $B(\underline{t})$ is given by:*

$$\lambda_1^\alpha(B(\underline{t})) = \begin{cases} \alpha^2 (G_1(\alpha t_1)^2 + \cdots + G_1(\alpha t_N)^2), & \alpha > 0, \\ -\alpha^2 (H_1(-\alpha t_1)^2 + \cdots + H_1(-\alpha t_N)^2), & \alpha < 0. \end{cases}$$

Proof. This follows by combining Lemma 2.1 and Remark 2.4. $\square$

Lemma 2.8. *For a box of fixed volume, the degenerate box (i.e., having at least one vanishing edge length)*

1. *minimises λ_1 for $\alpha < 0$,*
2. *maximises λ_1 for $\alpha > 0$.*

Proof. This is a direct consequence of Lemma 2.5. $\square$

Lemma 2.9. *If $f_1, \dots, f_N$ are strictly convex functions on $\mathbb{R}$ then*

$$|(f_1(z_1), \dots, f_N(z_N))|$$

is a strictly convex function of $z = (z_1, \dots, z_N) \in \mathbb{R}^N$

Proof. Let $z, w \in \mathbb{R}^N$ be any two vectors and $\tau \in (0, 1)$ parametrise the line joining them. Then:

$$\begin{aligned} & |(f_1((1-\tau)w_1 + \tau z_1), \dots, f_N((1-\tau)w_N + \tau z_N))| \\ & < |((1-\tau)f_1(w_1) + \tau f_1(z_1), \dots, (1-\tau)f_N(w_N) + \tau f_N(z_N))| \\ & \leq (1-\tau)|(f_1(w_1), \dots, f_N(w_N))| + \tau|(f_1(z_1), \dots, f_N(z_N))|, \end{aligned}$$

where the first inequality uses the strict convexity of the f_i and the second is the triangle inequality. $\square$

Lemma 2.10. *Provided the global minimum of a strictly convex function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ exists, it can be attained by at most one point [5].*

Proof. Suppose f is strictly convex i.e., for all $x, y \in \mathbb{R}^N$ and all $\tau \in (0, 1)$,

$$f(\tau x + (1-\tau)y) < \tau f(x) + (1-\tau)f(y).$$

Suppose further that f has two local minima, x_1 and x_2 and without loss of generality $f(x_1) \leq f(x_2)$. Then

$$f(\tau x_1 + (1-\tau)x_2) < \tau f(x_1) + (1-\tau)f(x_2) \leq \tau f(x_2) + (1-\tau)f(x_2) = f(x_2).$$

But this contradicts the fact that x_2 is a minimum since any neighbourhood of x_2 contains a point $\tau x_1 + (1-\tau)x_2$ when τ is sufficiently small. $\square$

Corollary 2.10.1. *Any local minimum of a strictly convex function is a global minimum.*

With all the necessary framework in place, we are ready to prove that the cube extremises the first Robin eigenvalue, *Theorem III.3* in Laugesen [25].

Theorem 2.11 (Laugesen, 2019). *Among all boxes of a fixed volume,*

1. *The cube minimises $\lambda_1^\alpha(B(\underline{t}))$ for $\alpha > 0$,*
2. *The cube maximises $\lambda_1^\alpha(B(\underline{t}))$ for $\alpha < 0$.*

Proof. We can embed the space of boxes in $\mathbb{R}^N$ by representing the box $B(\underline{t})$ that has dimensions $2t_1, \dots, 2t_N$ with the point $\underline{t} \in \mathbb{R}^N$. Since each $t_i > 0$ we can write $e^{\underline{z}} := (e^{z_1}, \dots, e^{z_N}) = (t_1, \dots, t_N)$. Then a box $B(e^{\underline{z}})$ with volume $V > 0$ satisfies

$$V = \prod_{i=1}^N 2e^{z_i} = 2^N \exp(z_1 + \dots + z_N) \Rightarrow \sum_{i=1}^N z_i = \log(2^{-N}V).$$

In our embedding in $\mathbb{R}^N$, the space of boxes of a fixed volume is represented by an $N-1$ dimensional subspace orthogonal to $(1, 1, \dots, 1)$, called $\Pi := \{\underline{z} \in \mathbb{R}^N : z_1 + \dots + z_N = \log(2^{-N}V)\}$ and cubes are represented by the line spanned by $(1, 1, \dots, 1)$. Let $\alpha > 0$. We first prove that $\sqrt{\lambda_1^\alpha(B(e^{\underline{z}}))}$ is a strictly convex function of $\underline{z} \in \mathbb{R}^N$. By Lemma 2.7,

$$\sqrt{\lambda_1^\alpha(B(e^{\underline{z}}))} = \alpha |(G_1(\alpha e^{z_1}), \dots, G_1(\alpha e^{z_N}))|.$$

By Lemma 2.6, $G_1(\alpha e^{z_j})$ is a convex function of z_j for $j = 1, \dots, N$ and Lemma 2.9 gives the desired result. By Remark 2.2, Π is convex and by Remark 2.3 the function $f(\underline{z}) = \sqrt{\lambda_1(B(e^{\underline{z}}), \alpha)}$ is strictly convex when restricted to Π . We want to show that f has a strict global minimum at the cube. Thanks to the strict convexity we can simply show that the gradient of f restricted to Π vanishes at the intersection point of Π with the line spanned by $(1, 1, \dots, 1)$. Then by Corollary 2.10.1, this must be the global minimum. This happens if ∇f is orthogonal to Π so is parallel to $(1, 1, \dots, 1)$. We can visualise this easily in 2 dimensions: a function has a stationary point when restricted to the line $x + y = 1$ if the gradient is parallel to $(1, 1)$ at the point $(1/2, 1/2)$. But ∇f must be parallel to $(1, 1, \dots, 1)$ at the intersection of Π with $\text{span}\{(1, 1, \dots, 1)\}$ due to the invariance of $f(z_1, \dots, z_N)$ under permutation of the z_i s. The case $\alpha < 0$ follows analogously. $\square$

2.3 The Equilateral Triangle

The eigenvalue problem on the triangle was first investigated by Lamé in 1833 when investigating the cooling of a triangular prism [23]. In doing so he found the spectrum of the Laplacian on an equilateral triangle, but provided no proof for his results. It was McCartin who made Lamé's results rigorous using more elementary methods. Below we follow the work of McCartin, beginning by introducing a triangular coordinate system. [31].

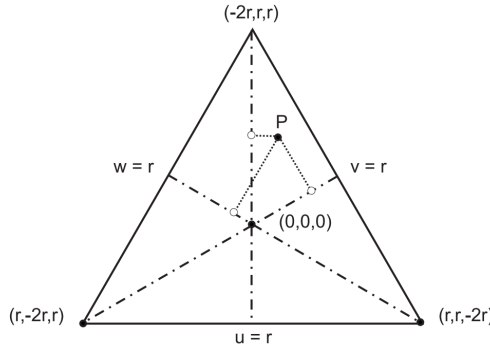


Figure 4: Triangular Coordinate System [31]

We can view the Cartesian coordinates of a point in $\mathbb{R}^N$ as measuring the distance from the origin to the projections of the point onto each of the axes, where we must pick a positive and negative direction. In a similar manner, we measure the distance from the centre of the triangle's incircle to the projection of a point onto each of the triangle's altitudes, measured positively towards an edge and negatively towards a vertex. If the triangle has edge length h then the radius of an inscribed circle, the *inradius*, is $r = \frac{h}{2\sqrt{3}}$. As the altitudes have length $3r$ and the centre is

the intersection of the concurrent altitudes, the centre has coordinates $(0,0,0)$. Thus each edge corresponds to one of the coordinates being r . For each vertex, two of the coordinates are r as it is the intersection of 2 edges, while the other coordinate is $-2r$. The explicit expressions for the coordinates are given by

$$u = r - y, \quad (2.6)$$

$$v = \frac{\sqrt{3}}{2} \left(x - \frac{h}{2} \right) - \frac{1}{2}(r - y), \quad (2.7)$$

$$w = \frac{\sqrt{3}}{2} \left(\frac{h}{2} - x \right) - \frac{1}{2}(r - y). \quad (2.8)$$

It is helpful to observe that

$$u + v + w = 0.$$

To rewrite the Laplacian in triangular coordinates we notice that

$$\frac{\partial^2}{\partial y^2} = \frac{\partial^2}{\partial u^2}$$

and

$$v - w = \sqrt{3} \left(x - \frac{h}{2} \right) \Rightarrow \frac{\partial^2}{\partial x^2} = 3 \frac{\partial^2}{\partial (v - w)^2}$$

so that

$$\nabla^2 = \frac{\partial^2}{\partial u^2} + 3 \frac{\partial^2}{\partial (v - w)^2}.$$

Then in seeking an eigenfunction we propose a separable ansatz, $f(u)g(v - w)$. That this is an eigenfunction implies

$$-f''g - 3fg'' = \lambda fg$$

and dividing by fg and using the standard separation of variables argument gives

$$\begin{cases} f'' + \zeta^2 f = 0, \\ g'' + \beta^2 g = 0, \end{cases}$$

subject to

$$\zeta^2 + 3\beta^2 = \lambda.$$

Thus f and g are trigonometric, or potentially hyperbolic if $\lambda < 0$. We next decompose $U = f(u)g(v - w)$ into symmetric and antisymmetric parts about the vertical altitude $v - w = 0$,

$$U_s(u, v, w) = \frac{U(u, v, w) + U(u, w, v)}{2}, \quad U_a(u, v, w) = \frac{U(u, v, w) - U(u, w, v)}{2}.$$

Let us take a brief interlude into the Neumann problem to provide a little inspiration for the form our eigenfunctions might take. The Dirichlet problem would serve just as well for this purpose. The symmetric Neumann eigenfunction, T_s , must satisfy $\frac{\partial T_s}{\partial u} = 0$ at the points $u = r$ and $u = -r$ so is given by

$$T_s = \cos\left(\frac{\pi l}{3r}(u + 2r)\right) \cos(\beta(v - w))$$

where $l \in \mathbb{Z}$ and $\frac{\pi^2 l^2}{9r^2} + 3\beta^2 = \lambda$. It was shown by McCartin in [27] for the Dirichlet case and in [28] for the Neumann case that one such term cannot satisfy the boundary condition and neither can a sum of two terms of this kind. However, a sum of three terms of this kind can. Hence we introduce a potential eigenfunction of the form

$$T_s = \sum_{i=1}^3 \cos\left(\frac{\pi l_i}{3r}(u + 2r)\right) \cos(\beta_i(v - w))$$

where $l_i \in \mathbb{Z}$ and

$$\frac{\pi^2 l_i^2}{9r^2} + 3\beta_i^2 = \lambda \quad (2.9)$$

for $i = 1, 2, 3$. By construction this satisfies the Neumann boundary condition on $u = r$. What does the Neumann boundary condition look like on $v = r$? We need not consider the edge $w = r$ thanks to the symmetry of our solution about $v = w$. We switch briefly to a coordinate system $(\sigma, \tau) = (w - u, w + u)$ along the line $v = r$. On this line $v - w = r - w = r + u + v = u + 2r$, using $u + v + w = 0$. Then the normal vector to the edge $v = r$ is parallel to the altitude described by $u = w$ meaning in (σ, τ) coordinates the unit normal is $(0, 1)$. Hence

$$\nabla T_s \cdot \boldsymbol{\nu}|_{v=r} = \frac{\partial T_s}{\partial \tau} \Big|_{v=r} = 0$$

is our Neumann condition on $v = r$. Our three term ansatz for T_s yields:

$$\begin{aligned} \frac{\partial T_s}{\partial(w+u)} \Big|_{v=r} = \sum_{i=1}^3 \left\{ \left(\frac{3}{4}\beta_i - \frac{\pi l_i}{12r} \right) \sin \left[\left(\frac{\pi l_i}{3r} + \beta_i \right) (u + 2r) \right] \right. \\ \left. + \left(\frac{3}{4}\beta_i + \frac{\pi l_i}{12r} \right) \sin \left[\left(-\frac{\pi l_i}{3r} + \beta_i \right) (u + 2r) \right] \right\} \end{aligned}$$

A strategic choice of l_i and β_i causes the above expression to vanish:

$$\frac{\pi l_1}{3r} + \beta_1 = -\frac{\pi l_3}{3r} + \beta_3, \quad (2.10a)$$

$$\frac{\pi l_1}{3r} - \beta_1 = -\frac{\pi l_2}{3r} - \beta_2, \quad (2.10b)$$

$$\frac{\pi l_2}{3r} - \beta_2 = -\frac{\pi l_3}{3r} - \beta_3. \quad (2.10c)$$

Adding these three equations together we get the crucial relation:

$$l_1 + l_2 + l_3 = 0. \quad (2.11)$$

Then adding equations (2.10a) and (2.10c) and using (2.11) in the second step gives:

$$\beta_1 - \beta_2 = -\frac{2\pi l_3}{3r} - \frac{\pi l_1}{3r} - \frac{\pi l_2}{3r} = \frac{\pi}{3r}(l_1 + l_2). \quad (2.12)$$

Then subtracting the the $i = 2$ instance of equation (2.9) from the $i = 1$ equation gives:

$$3\beta_1^2 - 3\beta_2^2 = \frac{\pi^2}{9r^2}(l_2^2 - l_1^2) \Rightarrow (\beta_1 - \beta_2)(\beta_1 + \beta_2) = \frac{\pi^2}{27r^2}(l_2 - l_1)(l_2 + l_1). \quad (2.13)$$

Comparing equations (2.12) and (2.13) we conclude that

$$\beta_1 + \beta_2 = \frac{\pi}{9r}(l_2 - l_1). \quad (2.14)$$

Adding equations (2.12) and (2.14) gives

$$\begin{aligned} 2\beta_1 &= \frac{\pi}{9r}(3l_1 + 3l_2 + l_2 - l_1) = \frac{\pi}{9r}(4l_2 + 2l_1) = \frac{\pi}{9r}(2(-l_3 - l_1) + 2l_2 + 2l_1) \\ &= \frac{2\pi}{9r}(l_2 - l_3). \end{aligned}$$

We can go through the same motions again and conclude that

$$\beta_1 = \frac{\pi(l_2 - l_3)}{9r}, \quad \beta_2 = \frac{\pi(l_3 - l_1)}{9r}, \quad \beta_3 = \frac{\pi(l_1 - l_2)}{9r}.$$

Having applied the Neumann condition to constrain the parameters β_i and l_i , the symmetric eigenfunction becomes

$$\begin{aligned} T_s^{l_2, l_3} = & \cos\left(\frac{\pi l_1}{3r}(u+2r)\right) \cos\left(\frac{\pi(l_2-l_3)}{9r}(v-w)\right) \\ & + \cos\left(\frac{\pi l_2}{3r}(u+2r)\right) \cos\left(\frac{\pi(l_3-l_1)}{9r}(v-w)\right) \\ & + \cos\left(\frac{\pi l_3}{3r}(u+2r)\right) \cos\left(\frac{\pi(l_1-l_2)}{9r}(v-w)\right). \end{aligned}$$

This bears remarkable resemblance to our computation on the box: despite the fact that we gave ourselves 3 degrees of freedom to begin with (l_1, l_2 , and l_3) we end up with a family of eigenfunctions being the sum of products of 2 trigonometric functions indexed by 2 integers. We can also compute the eigenvalues which are given by

$$\frac{2}{27} \left(\frac{\pi}{r}\right)^2 (l_1^2 + l_2^2 + l_3^2) = \frac{2}{27} \left(\frac{\pi}{r}\right)^2 ((-l_2-l_3)^2 + l_2^2 + l_3^2) = \frac{4}{27} \left(\frac{\pi}{r}\right)^2 (l_2^2 + l_2 l_3 + l_3^2).$$

Now we can go through the same steps for the antisymmetric eigenfunction by replacing $\cos(\beta_i(v-w))$ with $\sin(\beta_i(v-w))$ and we achieve the same eigenfunction with the relevant cosines replaced by sines and the same eigenvalues.

How do we adapt this to the Robin problem on the triangle? We provide an ansatz for the Robin eigenfunction on the triangle which now involves a linear combination of sine and cosine [29]. We manifest this linear combination by a shift in the argument of cosine. Once again we construct the symmetric eigenfunction:

$$\begin{aligned} U_s = & \cos\left(\frac{\pi\eta}{3r}(u+2r) - \delta_1\right) \cos\left(\frac{\pi(\mu-\nu)}{9r}(v-w)\right) \\ & + \cos\left(\frac{\pi\mu}{3r}(u+2r) - \delta_2\right) \cos\left(\frac{\pi(\nu-\eta)}{9r}(v-w)\right) \\ & + \cos\left(\frac{\pi\nu}{3r}(u+2r) - \delta_3\right) \cos\left(\frac{\pi(\eta-\mu)}{9r}(v-w)\right), \end{aligned}$$

where

$$\eta + \mu + \nu = 0.$$

This has eigenvalue:

$$\frac{2}{27} \left(\frac{\pi}{r}\right)^2 (\eta^2 + \mu^2 + \nu^2) = \frac{2}{27} \left(\frac{\pi}{r}\right)^2 ((-\mu-\nu)^2 + \mu^2 + \nu^2) = \frac{4}{27} \left(\frac{\pi}{r}\right)^2 (\mu^2 + \mu\nu + \nu^2).$$

Now we aim to find $\delta_1, \delta_2, \delta_3$ and η, μ, ν such that the Robin condition is satisfied along the boundary. Notice that η, μ, ν may indeed be complex in the negative α case as the first eigenvalue is negative and we expect hyperbolic eigenfunctions [30]. Imposing the Robin condition along $u = r$ gives:

$$\tan(\pi\eta - \delta_1) = \frac{3\alpha r}{\pi\eta}, \quad \tan(\pi\mu - \delta_2) = \frac{3\alpha r}{\pi\mu}, \quad \tan(\pi\nu - \delta_3) = \frac{3\alpha r}{\pi\nu}. \quad (2.15)$$

Note the resemblance to the function g_1 from the interval. Imposing now along $v = r$ and recalling that the symmetry of the eigenfunction means that the boundary condition must automatically be satisfied along $w = r$, we find

$$\tan\left(-\frac{\delta_2 + \delta_3}{2}\right) = \frac{3\alpha r}{\pi\eta}, \quad \tan\left(-\frac{\delta_1 + \delta_3}{2}\right) = \frac{3\alpha r}{\pi\mu}, \quad \tan\left(-\frac{\delta_1 + \delta_2}{2}\right) = \frac{3\alpha r}{\pi\nu}. \quad (2.16)$$

First let us define the auxilliary variables

$$L = -\frac{\delta_2 + \delta_3}{2}; \quad M = -\frac{\delta_1 + \delta_3}{2}; \quad N = -\frac{\delta_1 + \delta_2}{2}.$$

We can use these to re-express our sought-after parameters:

$$\delta_1 = L - M - N; \quad \delta_2 = -L + M - N; \quad \delta_3 = -L - M + N.$$

Then equating the corresponding terms in equations (2.15) and (2.16) we see that the arguments must differ by an integer multiple of π , so that from the second equation:

$$\pi\mu - \delta_2 = M + m\pi \Rightarrow \mu = \frac{1}{\pi}(2M - L - N) + m, \quad m \in \mathbb{Z}.$$

Similarly for the third equation:

$$\nu = \frac{1}{\pi}(2N - L - M) + n, \quad n \in \mathbb{Z}.$$

Lastly, using the constraint $\eta + \mu + \nu = 0$ we see that

$$\eta = -\mu - \nu = \frac{1}{\pi}(2L - N - M) - m - n.$$

Hence we can use the above expressions for η, μ and ν to rewrite the system (2.16) as

$$\begin{cases} (2L - M - N - (m + n)\pi) \tan(L) &= 3\alpha r, \\ (2M - L - N + n\pi) \tan(L) &= 3\alpha r, \\ (2N - M - L + m\pi) \tan(L) &= 3\alpha r. \end{cases}$$

This system can't be solved analytically but having been solved numerically in [29], Figure 5 supports the fact that we do in fact recover the Neumann and Dirichlet eigenfunctions respectively in the limits $\alpha \rightarrow 0^+$ and $\alpha \rightarrow \infty$.

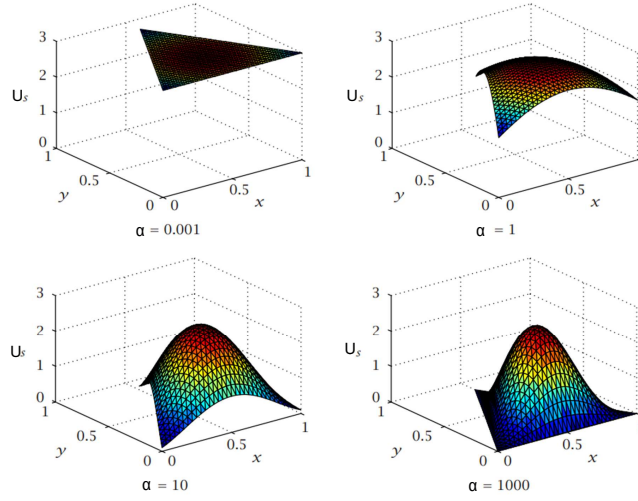


Figure 5: The First Neumann eigenfunction tends to the first Dirichlet eigenfunction as α runs from 0 to ∞ [29].

2.4 Local Optimisation of Polygonal Domains

Laugesen’s result is a global one, relying on the convexity of $\lambda_1^\alpha(B(t))$. However for more general domains this is too much to ask so we instead seek local minimisers. The idea of locality requires a notion of distance between shapes, which isn’t immediate. There are many different metrics one can define, however the *Hausdorff distance* is possibly the most intuitive and convenient. We follow the definitions of Henrot [16].

Definition 2.5. Let K_1, K_2 be compact sets contained in a larger ambient compact set $B \subset \mathbb{R}^N$. Then we define

1. $\rho(K_1, K_2) := \sup_{x \in K_1} d(x, K_2)$.

Notice this supremum is achieved as $d(\cdot, K_2)$ is a continuous function (vacuously) on a compact set.

2. $d^H(K_1, K_2) := \max\{\rho(K_1, K_2), \rho(K_2, K_1)\}$.

We can extend the definition to bounded, open subsets, $\Omega_1, \Omega_2 \subset B$, by defining the Hausdorff distance:

3. $d_H(\Omega_1, \Omega_2) := d^H(B \setminus \Omega_1, B \setminus \Omega_2)$.

The sets on the right-hand side are compact and non-empty (they contain ∂B).

Figure 6 gives an example of the Hausdorff distance between two sets. Having seen that the square is a global extremiser of the first eigenvalue among rectangles of fixed area, we might be inspired to imagine that an analogous result holds for the equilateral triangle. While this result holds in the Dirichlet case [35], it was only in 2023 that Krejčířík, Lotoreichik, and Vu proved that the equilateral triangle is a local maximiser of λ_1^α for $\alpha < 0$, subject to several restrictions [22].

Theorem 2.12 (Krejčířík, Lotoreichik, and Vu, 2023). *Let Ω^* be the equilateral triangle. Then there exists $\alpha_0 < 0$ depending only on $|\Omega^*|$ such that for all $\alpha \in [\alpha_0, 0)$,*

$$\lambda_1^\alpha(\Omega) \leq \lambda_1^\alpha(\Omega^*),$$

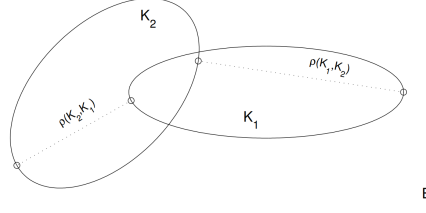


Figure 6: The Hausdorff distance between two sets [16].

for all triangles Ω with area equal to $|\Omega^*|$ and sufficiently close in Hausdorff distance.

The proof involves parametrising an arbitrary triangle of a fixed area, finding the eigenvalues on the triangle using a linear map onto the equilateral triangle and then differentiating the first eigenvalue with respect to each of the parameters. There are two parameters and both first derivatives vanish for Ω^* and the Hessian matrix is made negative-definite at Ω^* by imposing the above restriction on α . The proof relies on McCartin's expression for the eigenvalues on the equilateral triangle.

In 2024 Larsen-Scott and Clutterbuck used this method to broaden the class over which the square is a local maximiser of λ_1^α for $\alpha < 0$ [24].

Theorem 2.13 (Larsen-Scott and Clutterbuck, 2024). *Let $\alpha < 0$ and Ω^* be the square. Then*

$$\lambda_1^\alpha(\Omega) \leq \lambda_1^\alpha(\Omega^*),$$

for all quadrilaterals Ω with area $|\Omega^*|$ sufficiently close to Ω^* in Hausdorff distance.

The proof uses a similar argument, splitting a quadrilateral into 2 triangles and using a piecewise linear map to map an arbitrary quadrilateral onto the square [22]. The method then proceeds analogously, however without the constraints on the size of α .

2.5 The Disc

Polygonal domains are interesting in their own right however they quickly fall out of use in favour of discs when we relax the polygonal constraint in eigenvalue optimisation problems. Therefore it'll be useful to characterise the eigenvalues on the open ball in $\mathbb{R}^2$ of radius a . This example follows from Levitin, Mangoubi, and Polterovich [26]. We need the following expression for the Laplacian in polar coordinates:

$$\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial^2}{\partial \theta^2}.$$

Once again using a separable ansatz, $u = R(r)\Theta(\theta)$, gives:

$$-\nabla^2 R(r)\Theta(\theta) = -R''(r)\Theta(\theta) - \frac{1}{r}R'(r)\Theta(\theta) - \frac{1}{r^2}R(r)\Theta''(\theta) = \lambda R(r)\Theta(\theta).$$

Multiplying by $\frac{r^2}{R(r)\Theta(\theta)}$ yields:

$$-\frac{r^2 R''(r)}{R(r)} - \frac{r R'(r)}{R(r)} - \lambda r^2 = \frac{\Theta''(\theta)}{\Theta(\theta)}.$$

Arguing once again that each side must be constant and using the fact that $\Theta(\theta)$ is 2π -periodic, we conclude

$$\Theta(\theta) = A \sin(n\theta) + B \cos(n\theta), \quad n \in \mathbb{Z},$$

and

$$R''(r) + \frac{1}{r}R'(r) + \left(\lambda - \frac{n^2}{r^2}\right)R(r) = 0. \quad (2.17)$$

Using the substitution $s = \sqrt{\lambda}r$, equation (2.17) becomes

$$\lambda R''(s) + \lambda \frac{1}{s}R'(s) + \left(\lambda - \frac{\lambda n^2}{s^2}\right)R(s) = 0.$$

Dividing by λ this is the Bessel equation, which has only one solution which is non-singular at $r = 0$, the n -th Bessel function of the first kind, $J_n(s)$. Hence

$$R(r) = J_n(\sqrt{\lambda}r).$$

Then the Robin condition states

$$\nabla u(r, \theta) \cdot \boldsymbol{\nu} + \alpha u(r, \theta)|_{r=a} = 0.$$

On the boundary of the disc the normal vector $\boldsymbol{\nu} = \mathbf{e}_r$ so the eigenvalues λ_n are solutions to

$$R'(a) + \alpha R(a) = 0 \Rightarrow \sqrt{\lambda_n}J_n'(\sqrt{\lambda_n}a) + \alpha J_n(\sqrt{\lambda_n}a) = 0$$

and the corresponding eigenfunctions, indexed by $n \in \mathbb{N}$ are

$$u_n = R(r)\Theta(\theta) = J_n(\sqrt{\lambda_n}r) \sin(n(\theta + \phi)),$$

where $\phi \in [0, 2\pi)$ manifests an orientation of the axes. Once again we see that the eigenvalues form a countable sequence.

3 Preliminaries

We now exchange the concrete examples for an abstract unifying theory where eigenfunctions may not be explicitly found, however we can still say much about them. We do this through the spectral theory of certain operators in a Hilbert space. The definitions, theorems and proofs which appear below are as in section 6 of Evans [8]. I've aimed to provide insight and detail where the proofs of Evans lack explanation.

3.1 Spectral Theory for Hilbert Spaces

From now on H always refers to a real Hilbert space which has inner product $(\cdot, \cdot)$ and norm $\|\cdot\|$. Let's begin by recalling some definitions and facts about Hilbert spaces.

Definition 3.1. 1. A linear operator T is bounded if there exists $C > 0$ such that for all $x \in H$,

$$\|Tx\| \leq C\|x\|.$$

2. The smallest such C is called the operator norm of T , denoted

$$\|T\| := \inf\{C > 0 : \|Tx\| \leq C\|x\| \text{ for all } x \in H\}.$$

3. A bounded linear operator $f : H \rightarrow \mathbb{R}$ is called a bounded linear functional on H .

4. The set of bounded linear functionals on H is called the dual space of H , denoted H^* .

5. We denote the pairing of a functional $f \in H^*$ with an element $x \in H$ by

$$\langle f, x \rangle := f(x).$$

The dual space H^* is in fact a vector space over the same field as H . If H is a Hilbert space then H^* is already familiar to us.

Theorem 3.1 (Riesz Representation Theorem). *If H is a Hilbert space then its dual space, H^* , can be canonically identified with H . That is to say that for each $f \in H^*$ there is a unique $y \in H$ such that*

$$\langle f, x \rangle = (y, x)$$

for all $x \in H$.

Definition 3.2. Let H be a Hilbert space and

$$B : H \times H \rightarrow \mathbb{R}$$

be a bilinear form, that is to say B is linear in both arguments.

1. B is bounded if there exists $\mu > 0$ such that

$$|B[u, v]| \leq \mu\|u\|\|v\|,$$

for all $u, v \in H$.

2. B is coercive if there exists $\nu > 0$ such that

$$\nu\|u\|^2 \leq B[u, u],$$

for all $u \in H$.

An incredibly powerful theorem which guarantees uniqueness and existence of solutions is the following.

Theorem 3.2 (Lax-Milgram). *Let H be a Hilbert space and*

$$B : H \times H \rightarrow \mathbb{R}$$

be a bounded, coercive bilinear form. For each $f \in H^$ there exists a unique $u \in H$ such that*

$$B[u, v] = \langle f, v \rangle$$

for all $v \in H$.

Proof. We will prove this in the case that B is symmetric ($B[u, v] = B[v, u]$ for all $u, v \in H$) but the theorem holds for non-symmetric B , the proof being more involved. Firstly, the combination of boundedness and coerciveness implies that

$$\nu \|u\|^2 \leq B[u, u] \leq \mu \|u\|^2, \quad (3.1)$$

meaning that B is positive-definite. The assumption of symmetry promotes $B[u, v]$ from a bilinear form to an inner product on H . Define the norm induced by $B[\cdot, \cdot]$, by

$$\|u\|_B := \sqrt{B[u, u]}.$$

Then equation (3.1) asserts that the norms $\|\cdot\|$ and $\|\cdot\|_B$ are equivalent. It is easily proven that a Cauchy sequence converges in $\|\cdot\|$ if and only if it converges in $\|\cdot\|_B$, and so we endow H with the inner product $B[\cdot, \cdot]$, giving a Hilbert space to which we can apply the Riesz Representation Theorem, Theorem 3.1. This says that for every $f \in H^*$ there exists $u \in H$ such that for all $v \in H$, $f(v) = \langle f, v \rangle = B[u, v]$, precisely the statement of the theorem. $\square$

To be able to conclude more than we can about general bounded operators, we must stipulate further useful properties.

Definition 3.3. A bounded linear operator $T : H \rightarrow H$ is compact if every bounded sequence x_n in H has a subsequence x_{n_k} such that Tx_{n_k} converges.

We won't prove the following theorem but its corollary will prove incredibly useful.

Theorem 3.3 (The Fredholm Alternative). *Let $T : H \rightarrow H$ be a compact operator. Then*

1. $\mathcal{N}(I - T)$ is finite dimensional,
2. $\mathcal{R}(I - T)$ is closed,
3. $\mathcal{R}(I - T) = \mathcal{N}(I - T^*)^\perp$,
4. $\mathcal{N}(I - T) = \{0\} \Leftrightarrow \mathcal{R}(I - T) = H$
5. $\dim \mathcal{N}(I - T) = \dim \mathcal{N}(I - T^*)$.

The immediate application of this is the following dichotomy.

Corollary 3.3.1. *Precisely one of the two following possibilities occurs.*

- (i) *For each $f \in H$, the equation $u - Tu = f$ has a unique solution.*
- (ii) *The equation $u - Tu = 0$ has a non-trivial solution*

Proof of Corollary. If the kernel of $I - T$ is trivial then possibility (ii) is automatically excluded. Moreover $I - T$ is injective and part 4 of the theorem tells us that $I - T$ is also surjective, implying possibility (i).

Otherwise, the kernel of $I - T$ contains a non-trivial vector (possibility (ii)) and by part 4 of the theorem, there must exist an $f \in H \setminus \mathcal{R}(I - T)$, excluding possibility (i). $\square$

This is analogous to the finite dimensional case where a matrix A is either singular, whereby $A\mathbf{u} = 0$ has a non-trivial solution, or non-singular, whereby for each $\mathbf{v} \in \mathbb{R}^N$ there is a unique solution to $A\mathbf{u} = \mathbf{v}$.

In a finite dimensional vector space an eigenvector with eigenvalue η is a non-trivial solution to the equation $(T - \eta I)\mathbf{v} = 0$. Such a solution exists provided the map $T - \eta I$ is not a bijection. For $\eta \neq 0$ this is equivalent to $\mathbf{v} - \frac{1}{\eta}T\mathbf{v} = 0$ which is case (ii) in Corollary 3.3.1. We use this notion of an eigenvalue to generalise the definition to Hilbert spaces.

Definition 3.4. Let $T : H \rightarrow H$ be a bounded linear operator.

1. The resolvent of T , $\rho(T)$, is the set

$$\rho(T) = \{\eta \in \mathbb{R} : (T - \eta I) \text{ is bijective}\}.$$

2. The spectrum of T , $\sigma(T)$, is the complement of the resolvent in $\mathbb{R}$,

$$\sigma(T) := \mathbb{R} \setminus \rho(T).$$

3. The point spectrum of T , $\sigma_p(T) \subset \sigma(T)$, is the set of eigenvalues of T ,

$$\sigma_p(T) := \{\eta \in \mathbb{R} : T - \eta I \text{ is not injective}\}.$$

Below we will use η to denote an element of the spectrum and reserve λ for Robin eigenvalues.

For a linear operator on finite dimensional vector space, all elements of the spectrum are eigenvalues and these form a finite set, whose cardinality is bounded above by the dimension of the domain. However in an infinite dimensional vector space we can still make conclusions about spectra using the assumption of compactness.

Theorem 3.4 (Spectral Theorem for Compact Operators). *Let $T : H \rightarrow H$ be a compact operator on an infinite dimensional Hilbert space. Then*

1. $0 \in \sigma(T)$,
2. $\sigma(T) \setminus \{0\} = \sigma_p(T) \setminus \{0\}$,
3. *Either $\sigma(T) \setminus \{0\}$ is finite or $\sigma(T) \setminus \{0\}$ is a sequence tending to 0.*

Proof. 1. Suppose for contradiction that $0 \in \rho(T)$. Then T is a bijection and so $I = T^{-1} \circ T$ must be compact as the composition of a bounded and a compact operator. Indeed if $\{x_n\}_{n \in \mathbb{N}}$ is a bounded sequence in H then T maps it to a sequence with a convergent subsequence, say $\{x_{n_k}\}_{k \in \mathbb{N}}$. Then since T^{-1} is bounded and so continuous, it preserves the continuity of the subsequence. As H is infinite dimensional, it has a countable orthonormal set $\{e_n\}_{n \in \mathbb{N}}$, which is bounded. But no subsequence can be mapped to a convergent sequence under I : indeed for $m \neq n$,

$$\|e_n - e_m\|^2 = (e_n - e_m, e_n - e_m) = 2.$$

Therefore the identity operator is not compact, contradicting the fact $0 \in \rho(T)$.

2. Let $\eta \in \sigma(T) \setminus \{0\}$. If $\eta \notin \sigma_p(T)$ then T has no eigenvector with eigenvalue η so $\mathcal{N}(T - \eta I) = \{0\}$. By the Fredholm alternative, Theorem 3.3, $\mathcal{R}(T - \eta I) = H$ so $T - \eta I$ is a bijection meaning $\eta \in \rho(T)$.
3. Let η_k be a sequence of distinct elements in $\sigma(T) \setminus \{0\}$ converging to η . We would like to show that $\eta = 0$. As $\eta_k \in \sigma_p(T)$, there exists $w_k \neq 0$ such that $T w_k = \eta_k w_k$. Let $H_k := \text{span}\{w_1, \dots, w_k\} \subset H$ so that $H_{k+1} \setminus H_k$ is non-empty as the eigenvectors are linearly independent, given they correspond to distinct eigenvalues. Notice that $(T - \eta_k I)H_k \subset H_{k-1}$ as $(T - \eta_k I)w_k = 0$ so $T - \eta_k I$ removes any w_k terms from a linear combination in H_k . For

a fixed $k \in \mathbb{N}$ choose $u_k \in H_k \cap H_{k-1}^\perp$ and normalise so that $\|u_k\| = 1$. Now if $l < k$ then $H_{l-1} \subset H_l \subset H_{k-1} \subset H_k$. Thus

$$\begin{aligned} \left\| \frac{Tu_k}{\eta_k} - \frac{Tu_l}{\eta_l} \right\| &= \left\| \frac{Tu_k - \eta_k u_k}{\eta_k} - \frac{Tu_l - \eta_l u_l}{\eta_l} + u_k - u_l \right\| \\ &= \left\| \frac{Tu_k - \eta_k u_k}{\eta_k} - \frac{Tu_l - \eta_l u_l}{\eta_l} - u_l \right\| + \|u_k\| \\ &\geq 1 \end{aligned}$$

as $(T - \eta_k I)u_k, (T - \eta_l I)u_l, u_l$ all belong to H_{k-1} . If $\eta_n \rightarrow \eta \neq 0$ then $\frac{1}{\eta_n}u_n$ is a bounded sequence, so by compactness of T has a subsequence whose image under T converges. But this is in contradiction to the above which asserts that no such sequence exists. $\square$

Definition 3.5. If $T : H \rightarrow H$ is a bounded linear operator then its adjoint, T^* , satisfies

$$(Tx, y) = (x, T^*y)$$

for all $x, y \in H$. An operator is self-adjoint if $T = T^*$.

Such operators have some desirable qualities.

Lemma 3.5. *The eigenvalues of a self-adjoint operator are real.*

Proof. Suppose $Tx = \eta x$ for some $x \in H$ and some $\eta \in \mathbb{C}$. Then

$$(Tx, x) = (\eta x, x) = \eta(x, x) = (x, Tx) = (x, \eta x) = \bar{\eta}(x, x)$$

So $\eta = \bar{\eta} \Rightarrow \eta \in \mathbb{R}$. $\square$

Lemma 3.6. *Eigenfunctions of a self-adjoint operator are orthogonal.*

Proof. Suppose T is a self-adjoint operator with respect to $(\cdot, \cdot)$ and has eigenvectors $Tv_i = \eta_i v_i$ for $i = 1, 2$. Then

$$0 = (Tv_1, v_2) - (v_1, Tv_2) = (\eta_1 - \eta_2)(v_1, v_2)$$

and $\eta_1 \neq \eta_2$ gives the desired result. $\square$

This implies further that distinct eigenspaces are orthogonal.

The spectrum is actually a bounded set.

Lemma 3.7. *Let $T : H \rightarrow H$ be self-adjoint and bounded. Define*

$$\begin{aligned} m &= \inf\{(Tu, u) : u \in H, \|u\| = 1\}, \\ M &= \sup\{(Tu, u) : u \in H, \|u\| = 1\}. \end{aligned}$$

Then $\sigma(T) \subset [m, M]$ and $m, M \in \sigma(T)$.

Proof. Let $\eta > M$ and define a bilinear form on H

$$B_\eta[u, v] := (\eta u - Tu, v).$$

Then for all $u \in H$

$$B_\eta[u, u] = (\eta u - Tu, u) = \eta \|u\|^2 - (Tu, u) = \eta \|u\|^2 - \|u\|^2 \left(T \left(\frac{u}{\|u\|} \right), \frac{u}{\|u\|} \right) \geq (\eta - M) \|u\|^2. \quad (3.2)$$

Moreover, using the Cauchy-Schwarz inequality,

$$|B_\eta[u, v]| \leq \|\eta u - Tu\| \|v\| \leq (\eta + \|T\|) \|u\| \|v\|. \quad (3.3)$$

Equations (3.2) and (3.3) assert that B_η is bounded and coercive and we can invoke Lax-Milgram and the Riesz representation theorem to conclude that for each $f \in H$ there exists a unique $u \in H$ such that

$$B_\eta[u, v] = (\eta u - Tu, v) = (f, v)$$

for all $v \in H$. But this is equivalent to $\eta I - T : H \rightarrow H$ being a bijection so that $\eta \in \rho(T)$. Similarly $\eta \in \rho(T)$ if $\eta < m$.

Now we will show that $M \in \sigma(T)$. The bilinear form above, B_M , is positive definite and symmetric using the self-adjointness of T . Then the Cauchy-Schwarz inequality yields:

$$B_M[u, v]^2 \leq B_M[u, u]B_M[v, v] \Rightarrow |(Mu - Tu, v)| \leq \sqrt{(Mu - Tu, u)(Mv - Tv, v)}$$

for all $u, v \in H$. In particular

$$\|Mu - Tu\| \leq C\sqrt{(Mu - Tu, u)}.$$

Now let u_k be a sequence of unit vectors in H such that $(Tu_k, u_k) \rightarrow M$ as $k \rightarrow \infty$. Then by the above bound, $\|Mu - Tu\| \rightarrow 0$. If $M \in \rho(T)$ then $MI - T$ is invertible and so

$$u_k = (MI - T)^{-1}(MI - T)u_k.$$

But the right-hand side approaches 0 as $k \rightarrow \infty$, a contradiction so $MI - T$ must not be bijective. $\square$

As well as the eigenvalues we can characterise the eigenvectors.

Theorem 3.8. *Let H be a separable Hilbert space and $T : H \rightarrow H$ be self-adjoint and compact. Then there exists a countable orthonormal basis of H consisting of eigenvectors of T .*

Proof. Let η_k be the sequence of distinct eigenvalues of T excluding 0. Let $\eta_0 = 0$, $H_0 = \mathcal{N}(T)$ and $H_k = \mathcal{N}(T - \eta_k I)$, the latter of which has positive, finite dimension according to the Fredholm alternative. Recall from Lemma 3.6 that eigenspaces of distinct eigenvalues are orthogonal. Let $\tilde{H}$ be the smallest subspace of H which contains $H_0, H_1, \dots$ i.e., $\tilde{H}$ contains all *finite* linear combinations of eigenvectors of T . We would like to show that $\tilde{H}$ is dense in H . Acting on a finite linear combination of eigenvectors with T produces another finite linear combination of eigenvectors, meaning that $T(\tilde{H}) \subset \tilde{H}$. Also if $v \in \tilde{H}^\perp$ then for all $u \in \tilde{H}$,

$$(u, v) = 0 \Rightarrow (Tu, v) = (u, Tv) = 0,$$

as $Tu \in \tilde{H}$, implying that $T(\tilde{H}^\perp) \subset \tilde{H}^\perp$. Now the operator T is self-adjoint and compact by assumption so its restriction to $\tilde{H}^\perp$, $\tilde{T} := T|_{\tilde{H}^\perp}$, is too. If η_j is an eigenvalue of $\tilde{T}$ then it is an eigenvalue of T too and the corresponding eigenvector, $w_j \in H_j \subset \tilde{H}$. But $\tilde{T}$ can only have eigenvectors in $\tilde{H}^\perp$ and so $w_j \in \tilde{H} \cap \tilde{H}^\perp = \{0\}$, meaning $\sigma(\tilde{T}) = \{0\}$. But then by Lemma 3.7, $\sigma(\tilde{T})$ is contained in the interval $[m, M]$ and m and M are the maximum and minimum values attained by $(\tilde{T}u, u)$ so that $(\tilde{T}u, u) = 0$ for all $u \in \tilde{H}^\perp$. But then for all $u, v \in \tilde{H}^\perp$,

$$0 = (\tilde{T}(u + v), u + v) - (\tilde{T}u, u) - (\tilde{T}v, v) = 2(\tilde{T}u, v).$$

Thus $\tilde{T} = 0$, meaning

$$\tilde{H}^\perp \subset \mathcal{N}(T) = H_0 \subset \tilde{H}.$$

But once again $\tilde{H} \cap \tilde{H}^\perp \Rightarrow \tilde{H}^\perp = \{0\}$, meaning that $\tilde{H}$ is dense in H . As each eigenspace is finite dimensional, we can construct an orthonormal basis for each and the union of these forms an orthonormal basis for H as each eigenspace is orthogonal, where it is important that H is separable so that $H_0 = \mathcal{N}(T)$ has a countable orthonormal basis. $\square$

3.2 Sobolev Spaces

In order to apply the spectral theory of Hilbert spaces, we need an appropriate Hilbert space in which to cast the eigenvalue problem. We aim to broaden the space in which we seek eigenfunctions from $C^2(\Omega)$. This section follows the conventions of Evans [8].

Definition 3.6. Denote the space of infinitely differentiable functions which are compactly supported in $\Omega \subset \mathbb{R}^N$ by $C_c^\infty(\Omega)$. Such a function is called a test function.

In order to motivate the following definitions, we take the Helmholtz equation, multiply it by a test function $\varphi \in C_c^\infty(\Omega)$ and integrate over Ω to find:

$$\begin{aligned} \int_{\Omega} \lambda \varphi u \, dx &= - \int_{\Omega} \varphi \nabla^2 u \, dx \\ &= \int_{\Omega} \nabla \varphi \cdot \nabla u \, dx - \int_{\partial\Omega} \varphi \frac{\partial u}{\partial \nu} \, d\sigma \\ &= - \int_{\Omega} u \nabla^2 \varphi \, dx + \int_{\partial\Omega} u \frac{\partial \varphi}{\partial \nu} \, d\sigma \\ &\Rightarrow \int_{\Omega} u (\nabla^2 \varphi + \lambda \varphi) \, dx = 0. \end{aligned} \tag{3.4}$$

We've integrated by parts in the second and third equalities and discarded the boundary terms using the fact that a test function's partial derivatives of all orders vanish on the boundary. A solution to (1.4) satisfies (3.4) for all test functions $\varphi \in C_c^\infty(\Omega)$, however we can use this formulation to loosen the sense in which we ask u to be an eigenfunction of the Laplacian. Classically, we require u to be twice differentiable inside Ω and continuous up to the boundary (i.e., $u \in C^2(\Omega) \cap C(\bar{\Omega})$) however we can expand the space of admissible solutions by weakening the differentiability we require.

Definition 3.7. Suppose $u, v \in L_{\text{loc}}^1(\Omega)$ and β is a multiindex. Then v is the β -th weak partial derivative of u , written $D^\beta u = v$, provided:

$$\int_{\Omega} u D^\beta \varphi \, dx = (-1)^{|\beta|} \int_{\Omega} v \varphi \, dx$$

for all test functions $\varphi \in C_c^\infty(\Omega)$.

Everywhere below $k \in \mathbb{N}$ and $1 \leq p < \infty$.

Definition 3.8. The Sobolev Space, denoted $W^{k,p}(\Omega)$, is the space of functions $u : \Omega \rightarrow \mathbb{R}$ for which the weak derivative $D^\beta u$ exists and belongs to $L^p(\Omega)$ for all multiindices β with $|\beta| \leq k$.

We see that $W^{0,p}(\Omega) = L^p(\Omega)$ and like L^p , $W^{k,p}(\Omega)$ is a Banach space for all p with norm

$$\|u\|_{W^{k,p}(\Omega)} = \left(\sum_{|\beta| \leq k} \|D^\beta u\|_{L^p(\Omega)}^p \right)^{1/p}.$$

Again similar to the L^p case, $W^{k,2}(\Omega)$ is a Hilbert space which we denote $H^k(\Omega)$ whose norm is induced by the inner product

$$\langle u, v \rangle_{H^k(\Omega)} := \sum_{|\beta| \leq k} (D^\beta u, D^\beta v)_{L^2(\Omega)}.$$

The Sobolev spaces contain functions that aren't classically differentiable or even continuous but thankfully we have the following approximation theorem to get a handle on how pathological they can be.

Theorem 3.9. Suppose $\partial\Omega$ is C^1 and $u \in W^{k,p}(\Omega)$. Then there exists a sequence of functions $u_n \in C^\infty(\bar{\Omega})$ such that

$$\|u - u_n\|_{W^{k,p}(\Omega)} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

We will focus almost exclusively on $H^1(\Omega)$ and make ample use of the inequality

$$\|u\|_{L^2(\Omega)} \leq \|u\|_{H^1(\Omega)}.$$

One problem we encounter is that by expanding our space of solutions to ensure existence, we have lost uniqueness. The boundary $\partial\Omega$ has Lebesgue measure zero and in $L^p(\Omega)$ and $W^{k,p}(\Omega)$ we define an equivalence relation between functions that agree everywhere spare a set of measure zero. Therefore an eigenfunction inside Ω is equal (modulo the equivalence relation) to another solution that takes a different value on the entire boundary. To rectify this we need to use the *trace operator* to define what it means for a function to satisfy the boundary condition in the weak sense.

Theorem 3.10 (The Trace Theorem). Assume that $\partial\Omega$ is C^1 . Then there exists a bounded linear operator

$$Tr : W^{1,p}(\Omega) \rightarrow L^p(\partial\Omega)$$

such that

$$Tr(u) = u|_{\partial\Omega} \text{ if } u \in W^{1,p} \cap C(\bar{\Omega})$$

and there exists a constant C independent of u such that

$$\|Tr(u)\|_{L^p(\partial\Omega)} \leq C\|u\|_{W^{1,p}(\Omega)},$$

for all $u \in W^{1,p}(\Omega)$.

The utility of this theorem is the fact that $C(\bar{\Omega})$ is dense in $W^{1,p}$ (by Theorem 3.9) and the continuity of Tr (equivalent to boundedness) allows the trace to be uniquely extended to all of $W^{1,p}$. The inequality (3.10) will be incredibly useful when discussing the boundary values of a H^1 function.

Definition 3.9. The trace operator is the map Tr stated to exist in Theorem 3.10.

From now on when we ask that a function in $H^1(\Omega)$ satisfies the Robin boundary condition, or any boundary condition for that matter, we ask that it does so in the sense of traces.

3.3 Spectral Theory of Elliptic Operators

Now we have a space in which Laplacian eigenfunctions might live, we can apply the spectral theory of Hilbert spaces. We still follow Evans [8] which is geared towards general elliptic operators obeying the Dirichlet boundary condition and I have worked to adapt this to the Robin condition.

Definition 3.10. An elliptic operator takes the form

$$\mathcal{L}u = - \sum_{i,j} a_{i,j} \partial_{x_i} \partial_{x_j} u + \sum_i b_i \partial_{x_i} u + cu,$$

where the coefficients are continuous and $a_{i,j}$ satisfies the ellipticity condition: that there exists a $K > 0$ such that

$$\sum_{i,j} a_{i,j}(x) \zeta_i \zeta_j \geq K|\zeta|^2, \quad (3.5)$$

for almost all $x \in \Omega$ and all $\zeta \in \mathbb{R}^N$. That is to say that the matrix with entries $a_{i,j}$ is positive definite.

The Laplacian is clearly expressible in this form where

$$a_{i,j} = \delta_{i,j}, \quad b_i = 0, \quad c = 0,$$

and it is elliptic as

$$\sum_{i,j} \delta_{i,j}(x) \zeta_i \zeta_j = \sum_i \zeta_i \zeta_i = |\zeta|^2.$$

Everything we will say below is proved for general elliptic operators with Dirichlet boundary conditions by Evans, however to adapt this to the Robin condition I have restricted my attention to the Laplacian. To ensure functions satisfy the Robin condition, I define the domain of my operators as follows.

Definition 3.11. Let $H_\alpha^1(\Omega)$ be the subset of $H^1(\Omega)$ satisfying the Robin condition in the sense of traces:

$$H_\alpha^1(\Omega) := \left\{ u \in H^1(\Omega) : \frac{\partial u}{\partial \nu} + \alpha u = 0 \text{ on } \partial\Omega \right\}.$$

We ought to check that this subset is indeed a Hilbert space.

Lemma 3.11. $H_\alpha^1(\Omega)$ is a Hilbert space with the inner product inherited from $H^1(\Omega)$.

The proof below, and the subsequent adaptation to the Robin condition is original.

Proof. We first show that $H_\alpha^1(\Omega)$ is a subspace of $H^1(\Omega)$: indeed if $u, v \in H_\alpha^1(\Omega)$ then

$$\frac{\partial(u+v)}{\partial \nu} + \alpha(u+v) = \left(\frac{\partial u}{\partial \nu} + \alpha u \right) + \left(\frac{\partial v}{\partial \nu} + \alpha v \right) = 0$$

and closure under scalar multiplication follows similarly. Being a subspace of a Hilbert space, to establish that $H_\alpha^1(\Omega)$ is a Hilbert space we need only show that it is closed. Indeed let H be a Hilbert space and $\hat{H} \subset H$ a subspace. Then a Cauchy sequence in $\hat{H}$, $\{x_n\}_{n \in \mathbb{N}}$, converges with limit in H as $\{x_n\}_{n \in \mathbb{N}}$ is a Cauchy sequence in the Hilbert space H and this limit is contained in $\hat{H}$ if $\hat{H}$ is a closed subspace.

To establish that $H_\alpha^1(\Omega)$ is closed, let $\{u_n\}_{n \in \mathbb{N}}$ be a sequence in $H_\alpha^1(\Omega)$ converging to u . We must show that u obeys the Robin condition:

$$\begin{aligned} \left\| \frac{\partial u}{\partial \nu} + \alpha u \right\|_{L^2(\partial\Omega)} &\leq \left\| \frac{\partial u - u_n}{\partial \nu} + \alpha(u - u_n) \right\|_{L^2(\partial\Omega)} + \left\| \frac{\partial u_n}{\partial \nu} + \alpha u_n \right\|_{L^2(\partial\Omega)} \\ &\leq \left\| \frac{\partial u - u_n}{\partial \nu} \right\|_{L^2(\partial\Omega)} + |\alpha| \|u - u_n\|_{L^2(\partial\Omega)} \\ &\leq C \left(\left\| \frac{\partial u - u_n}{\partial \nu} \right\|_{H^1(\Omega)} + |\alpha| \|u - u_n\|_{H^1(\Omega)} \right) \\ &= C \|\nu \cdot \nabla(u - u_n)\|_{H^1(\Omega)} + C|\alpha| \|u - u_n\|_{H^1(\Omega)} \\ &\leq C \|\nabla(u - u_n)\|_{H^1(\Omega)} + C|\alpha| \|u - u_n\|_{H^1(\Omega)}. \end{aligned}$$

Above we have used the Cauchy-Schwarz inequality and the boundedness of the trace operator from Theorem 3.10: that there exists a constant C independent of u such that

$$\|Tr(u)\|_{L^p(\partial\Omega)} \leq C \|u\|_{W^{1,p}(\Omega)},$$

for all $u \in W^{1,p}(\Omega)$. As the left-hand side is independent of n and the right-hand side can be made arbitrarily small by taking n sufficiently large, we conclude that u obeys the Robin condition. $\square$

To maintain the sense that this can be generalised I retain the notation L for the Laplacian herein, to remind that the proofs can be adapted, with no real change in structure, to general elliptic operators.

$$L = - \sum_i \partial_{x_i x_i}^2, \quad \mathcal{D}(L) = H_\alpha^1(\Omega).$$

We diverge from the study of the eigenfunctions of the Robin Laplacian and instead study L as an operator on the Sobolev space $H_\alpha^1(\Omega)$, considering first the Poisson problem:

$$\begin{cases} Lu = f & \text{in } \Omega, \\ \frac{\partial u}{\partial \nu} + \alpha u = 0 & \text{on } \partial\Omega. \end{cases} \quad (3.6)$$

Suppose u is a solution to (3.6) and $v \in H_\alpha^1(\Omega)$ is any trial function. Multiplying Poisson's equation by v and integrating over Ω , we find:

$$\begin{aligned} \int_\Omega f v \, dx &= \int_\Omega -v \nabla^2 u \, dx \\ &= \int_\Omega \nabla u \cdot \nabla v \, dx - \int_{\partial\Omega} v \frac{\partial u}{\partial \nu} \, d\sigma \\ &= \int_\Omega \nabla u \cdot \nabla v \, dx + \alpha \int_{\partial\Omega} uv \, d\sigma. \end{aligned}$$

The left-hand side is the $L^2(\Omega)$ inner product of f and v and the we give the following definition for the right-hand side.

Definition 3.12. The bilinear form associated to the Robin Laplacian is given by

$$B[u, v] := \int_\Omega \nabla u \cdot \nabla v \, dx + \alpha \int_{\partial\Omega} uv \, d\sigma.$$

We see that a classical $(C^2(\Omega) \cap C(\overline{\Omega}))$ solution of the Poisson problem (3.6) obeys $B[u, v] = (f, v)$ for all $v \in H_\alpha^1(\Omega)$, however this condition poses weaker constraints on u : namely that $u \in H_\alpha^1(\Omega)$. In particular u need only be once weakly differentiable. This motivates us to weaken the sense in which we ask u to be a solution of the Poisson problem.

Definition 3.13. We call $u \in H_\alpha^1(\Omega)$ a weak solution of the Robin problem if

$$B[u, v] = (f, v)$$

for all $v \in H_\alpha^1(\Omega)$ where $(\cdot, \cdot)$ is the inner product on $L^2(\Omega)$.

Lemma 3.12. *The Robin Laplacian is self-adjoint with respect to the L^2 inner product.*

Proof. Using integration by parts twice along with the Robin boundary condition we see

$$\begin{aligned} (Lu, v) &= \int_\Omega -\nabla^2 uv \, dx \\ &= \int_\Omega \nabla u \cdot \nabla v \, dx - \int_{\partial\Omega} v \frac{\partial u}{\partial \nu} \, d\sigma \\ &= \int_\Omega -u \nabla^2 v \, dx - \int_{\partial\Omega} v \frac{\partial u}{\partial \nu} \, d\sigma + \int_{\partial\Omega} u \frac{\partial v}{\partial \nu} \, d\sigma \\ &= \int_\Omega -u \nabla^2 v \, dx - \int_{\partial\Omega} v(-\alpha u) \, d\sigma + \int_{\partial\Omega} u(-\alpha v) \, d\sigma \\ &= \int_\Omega -u \nabla^2 v \, dx = (u, Lv). \end{aligned}$$

□

Now we can apply Lemmas 3.5 and 3.6 to the spectrum of the Robin Laplacian. We would like to apply the Lax-Milgram theorem (Theorem 3.2), which gives conditions on B such that a unique weak solution exists, to the Robin Laplacian, the legitimacy of which is the content of the following theorem.

Theorem 3.13. *Let $B[\cdot, \cdot]$ be the bilinear form associated to the Robin Laplacian defined above with $\alpha > 0$ from now on. Then there exist $\mu, \nu > 0$ and $\rho \geq 0$ such that*

$$1. |B[u, v]| \leq \mu \|u\|_{H_\alpha^1(\Omega)} \|v\|_{H_\alpha^1(\Omega)},$$

and

$$2. \nu \|u\|_{H_\alpha^1(\Omega)}^2 \leq B[u, u] + \rho \|u\|_{L^2(\Omega)}^2.$$

for all $u, v \in H_\alpha^1(\Omega)$.

Proof. 1. For boundedness we check

$$\begin{aligned} |B[u, v]| &\leq \left| \int_\Omega \nabla u \cdot \nabla v \, dx \right| + \left| \alpha \int_{\partial\Omega} uv \, d\sigma \right| \\ &\leq \left(\int_\Omega |\nabla u|^2 \, dx \right)^{1/2} \left(\int_\Omega |\nabla v|^2 \, dx \right)^{1/2} + |\alpha| \left(\int_{\partial\Omega} u^2 \, d\sigma \right)^{1/2} \left(\int_{\partial\Omega} v^2 \, d\sigma \right)^{1/2} \\ &= \|\nabla u\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)} + |\alpha| \|u\|_{L^2(\partial\Omega)} \|v\|_{L^2(\partial\Omega)} \\ &\leq \|\nabla u\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)} + C^2 |\alpha| \|u\|_{H_\alpha^1(\Omega)} \|v\|_{H_\alpha^1(\Omega)} \\ &\leq \mu \left(\int_\Omega |\nabla u|^2 + u^2 \, dx \right)^{1/2} \left(\int_\Omega |\nabla v|^2 + v^2 \, dx \right)^{1/2} \\ &= \mu \|u\|_{H_\alpha^1(\Omega)} \|v\|_{H_\alpha^1(\Omega)} \end{aligned}$$

where we have used the trace inequality (3.10) and in the final step we require that $\mu > C^2 |\alpha| + 1 > 0$.

2. Now for coerciveness. We will need to use Maz'ja's inequality [19] which states that there exists $c > 0$ such that

$$\|u\|_{L^2(\Omega)} \leq c \left(\|\nabla u\|_{L^2(\Omega)}^2 + \|u\|_{L^2(\partial\Omega)}^2 \right)$$

for all $u \in H_\alpha^1(\Omega)$. We will pre-emptively rearrange this as

$$-\alpha \|u\|_{L^2(\partial\Omega)}^2 \leq -\frac{\alpha}{c} \|u\|_{L^2(\Omega)}^2 + \alpha \|\nabla u\|_{L^2(\Omega)}^2.$$

Notice as well that c can be made larger if necessary, specifically it can be made larger than α . Then

$$\begin{aligned} \|u\|_{H_\alpha^1(\Omega)}^2 &= \int_\Omega |\nabla u|^2 + u^2 \, dx \\ &= B[u, u] - \alpha \|u\|_{L^2(\partial\Omega)}^2 + \|u\|_{L^2(\Omega)}^2 \\ &\leq B[u, u] + \|u\|_{L^2(\Omega)}^2 + \alpha \|\nabla u\|_{L^2(\Omega)}^2 - \frac{\alpha}{c} \|u\|_{L^2(\Omega)}^2 \\ &\leq B[u, u] \left(1 - \frac{\alpha}{c} \right) + \alpha B[u, u] \\ &= (1 + \alpha) B[u, u] + \left(1 - \frac{\alpha}{c} \right) \|u\|_{L^2(\Omega)}^2. \end{aligned}$$

Dividing by $1 + \alpha > 0$ gives the desired form of inequality.

This completes the proof. $\square$

We've encountered a small issue: if $\rho \neq 0$ then the conditions of the Lax-Milgram theorem are not precisely fulfilled. We can partially remedy this however with the following theorem.

Theorem 3.14. *If ρ is as in 3.13, for each $\eta \geq \rho$ and each $f \in L^2(\Omega)$ there is a unique weak solution $u \in H_\alpha^1(\Omega)$ to the problem*

$$\begin{cases} Lu + \eta u = f & \text{in } \Omega, \\ \frac{\partial u}{\partial \nu} + \alpha u = 0 & \text{on } \partial\Omega. \end{cases}$$

Proof. Let $\eta \geq \rho$ and define the new bilinear form:

$$B_\eta[u, v] := B[u, v] + \eta(u, v),$$

for all $u, v \in H_\alpha^1(\Omega)$ and $(\cdot, \cdot)$ is as ever the $L^2(\Omega)$ inner product. Then $B_\eta[\cdot, \cdot]$ satisfies the conditions of the Lax-Milgram theorem: boundedness follows since

$$\begin{aligned} |B_\eta[u, v]| &\leq |B[u, v]| + \eta|(u, v)| \\ &\leq \mu\|u\|_{H_\alpha^1(\Omega)}\|v\|_{H_\alpha^1(\Omega)} + \eta\|u\|_{L^2(\Omega)}\|v\|_{L^2(\Omega)} \\ &\leq \mu\|u\|_{H_\alpha^1(\Omega)}\|v\|_{H_\alpha^1(\Omega)} + \eta\|u\|_{H_\alpha^1(\Omega)}\|v\|_{H_\alpha^1(\Omega)} \\ &= (\mu + \eta)\|u\|_{H_\alpha^1(\Omega)}\|v\|_{H_\alpha^1(\Omega)}. \end{aligned}$$

And coerciveness:

$$\nu\|u\|_{H_\alpha^1(\Omega)}^2 \leq B[u, u] + \rho\|u\|_{L^2(\Omega)}^2 \leq B[u, u] + \eta\|u\|_{L^2(\Omega)}^2 = B_\eta[u, u]. \quad (3.7)$$

□

Theorem 3.15 (The Fredholm Alternative). *Precisely one of the following two possibilities occurs:*

1. *For each $f \in L^2(\Omega)$ there is a unique weak solution to*

$$\begin{cases} Lu = f & \text{in } \Omega, \\ \frac{\partial u}{\partial \nu} + \alpha u = 0 & \text{on } \partial\Omega. \end{cases} \quad (3.8)$$

2. *There exists a non-zero weak solution to*

$$\begin{cases} Lu = 0 & \text{in } \Omega, \\ \frac{\partial u}{\partial \nu} + \alpha u = 0 & \text{on } \partial\Omega. \end{cases} \quad (3.9)$$

Proof. Take $\eta = \rho$ as in Theorem 3.13 and define:

$$B_\rho[u, v] := B[u, v] + \rho(u, v)$$

which corresponds to the operator $L_\rho : H_\alpha^1(\Omega) \rightarrow L^2(\Omega), u \mapsto Lu + \rho u$. Then by Theorem 3.14, for each $g \in L^2(\Omega)$, there exists a unique $u \in H_\alpha^1(\Omega)$ such that $B_\rho[u, v] = (g, v)$ for all $v \in H_\alpha^1(\Omega)$. As u is unique, for each $g \in L^2(\Omega)$ we can formally define

$$L_\rho^{-1}g : L^2(\Omega) \rightarrow H_\alpha^1(\Omega).$$

as

$$u =: L_\rho^{-1}g$$

whenever u is the unique weak solution of $Lu + \rho u = g$. Now notice that u is a weak solution of (3.8) provided

$$B_\rho[u, v] = (\rho u + f, v)$$

for all $v \in H_\alpha^1(\Omega)$. That is, if

$$u = L_\rho^{-1}(\rho u + f) = \rho L_\rho^{-1}u + L_\rho^{-1}f. \quad (3.10)$$

Let us define

$$Ku := \rho L_\rho^{-1} u, \quad h = L_\rho^{-1} f.$$

Then we can rewrite (3.10) as

$$u - Ku = h.$$

We now claim that $K : L^2(\Omega) \rightarrow H_\alpha^1(\Omega) \subset L^2(\Omega)$ is linear, bounded, and compact in the hopes that we can invoke the Fredholm alternative. By the coerciveness of B_ρ ,

$$\nu \|u\|_{H_\alpha^1(\Omega)}^2 \leq B_\rho[u, u] = (g, u) \leq \|g\|_{L^2(\Omega)} \|u\|_{L^2(\Omega)} \leq \|g\|_{L^2(\Omega)} \|u\|_{H_\alpha^1(\Omega)},$$

implying that

$$\|u\|_{H_\alpha^1(\Omega)} \leq \frac{1}{\nu} \|g\|_{L^2(\Omega)}.$$

Now aiming to show that K is bounded,

$$\|Kg\|_{L^2(\Omega)} \leq \|Kg\|_{H_\alpha^1(\Omega)} = \|\rho L_\rho^{-1} g\|_{H_\alpha^1(\Omega)} = \rho \|u\|_{H_\alpha^1(\Omega)} \leq \frac{\rho}{\nu} \|g\|_{L^2(\Omega)}.$$

The compactness of K follows from the Rellich-Kondrachov theorem. Now we can use the Fredholm alternative, Theorem 3.3, to conclude that either

1. For each $h \in L^2(\Omega)$,

$$u - Ku = h$$

has a unique solution $u \in L^2(\Omega)$, otherwise

2. the equation

$$u - Ku = 0$$

has non-zero solutions in $L^2(\Omega)$.

□

Theorem 3.16 (Spectral Theorem for Elliptic Operators).

1. There exists a countable set $\sigma(L) \subset \mathbb{R}$ such that

$$\begin{cases} Lu = \lambda u + f & \text{in } \Omega, \\ \frac{\partial u}{\partial \nu} + \alpha u = 0 & \text{on } \partial\Omega. \end{cases} \quad (3.11)$$

has a unique weak solution for each $f \in L^2(\Omega)$ if and only if $\lambda \notin \sigma(L)$.

2. The eigenvalues of the Laplacian form a real, countable sequence, bounded from below and tending to ∞ .

3. The corresponding eigenfunctions form a basis of $L^2(\Omega)$.

Proof. Let ρ be as defined in Theorem 3.13. We can freely assume that $\rho > 0$ (if ρ satisfies the coerciveness inequality then any larger value must too) and let $\lambda > -\rho$. According to Theorem 3.15, the problem (3.11) has a unique weak solution if and only if $u \equiv 0$ is the only solution to the homogeneous problem

$$\begin{cases} Lu = \lambda u & \text{in } \Omega, \\ \frac{\partial u}{\partial \nu} + \alpha u = 0 & \text{on } \partial\Omega. \end{cases}$$

This is equivalent to $u \equiv 0$ being the only solution to

$$\begin{cases} Lu + \rho u = (\rho + \lambda)u & \text{in } \Omega, \\ \frac{\partial u}{\partial \nu} + \alpha u = 0 & \text{on } \partial\Omega. \end{cases}$$

This is solved by u if

$$u = L_\rho^{-1}(\rho + \lambda)u = \frac{\rho + \lambda}{\rho}Ku,$$

where $Ku = \rho L_\rho^{-1}u$ is a compact, linear operator as before. If $u \equiv 0$ is the only solution to

$$Ku = \frac{\rho}{\rho + \lambda}u$$

then $\frac{\rho}{\rho + \lambda}$ is not an eigenvalue on K , which is equivalent to equation (3.11) having a unique solution. As K is compact, according to Theorem 3.4, the eigenvalues of K form either a finite set or a sequence approaching zero. In the latter case, the problem (3.11) has a unique solution for all λ spare a sequence $\lambda_k \rightarrow \infty$. We now examine the formal inverse of the Laplacian with the view to applying the spectral theorem. The operator

$$S := L_0^{-1}$$

is bounded, linear, compact operator mapping $L^2(\Omega)$ to itself. We claim that S is self-adjoint. Let $f, g \in L^2(\Omega)$ so that $u := Sf$ implies that u is the weak solution to $Lu = f$ and likewise $v := Sg$ implies that v is the weak solution to $Lv = g$. Thus

$$(Sf, g) = (u, g) = B[u, v]$$

and

$$(f, Sg) = (f, v) = B[v, u].$$

So S is self-adjoint thanks to the symmetry of B . Furthermore, $(Sf, f) = (u, f) = B[u, u] \geq 0$ so we can apply the theory we developed of compact, self-adjoint operators. Lastly, if w is an eigenvector of S with corresponding eigenvalue η , that is $Sw = \eta w$, then applying L to both sides we see that $\frac{1}{\eta}w = Lw$ so that the sequence of eigenvalues tends to infinity. $\square$

We finish by including a theorem which ensures the regularity of weak solutions.

Theorem 3.17. *Suppose $a_{i,j}$, $b_{i,c}$, and $f \in C^\infty(\bar{\Omega})$. If $u \in H^1(\Omega)$ is a weak solution of*

$$\begin{cases} Lu = f \text{ in } \Omega, \\ \frac{\partial u}{\partial \nu} + \alpha u = 0 \text{ on } \partial\Omega, \end{cases}$$

then, provided $\partial\Omega$ is C^∞ , $u \in C^\infty(\bar{\Omega})$.

3.4 The Symmetric Decreasing Rearrangement

In the final section we will investigate functions on the ball in $\mathbb{R}^N$ which are spherically symmetric. In a sense we would like to build a spherically symmetric function out of a given function on an arbitrary domain which preserves its size. This section follows the lecture notes of Burchard [4], however the diagrams and the code used to generate them is original. We first fix some notation. We use $\mathbb{1}_A(x)$ to denote the indicator function for a set $A \subset \Omega$

$$\mathbb{1}_A(x) := \begin{cases} 1, & x \in A \\ 0, & x \notin A. \end{cases}$$

and the level sets of a non-negative L^2 function on Ω are given by

$$U_t = \{x \in \Omega : f(x) > t\}.$$

Using these two definitions we can represent a non-negative $L^2(\Omega)$ function using the layer cake decomposition

$$f(x) = \int_0^\infty \mathbb{1}_{U_t}(x) dt.$$

Now in the spirit of creating a spherically symmetric version of a function, we first do the same for sets.

Definition 3.14. Let A be a measurable set of finite, positive volume in $\mathbb{R}^N$. Its symmetric rearrangement A^* is the open ball with volume equal to that of A .

$$A^* := \{x \in \mathbb{R}^N : \omega_N |x|^N < |A|\},$$

where ω_N is the volume of the unit ball in $\mathbb{R}^N$.

We can thus build a function from the layer cake decomposition of f by symmetrising the level sets.

Definition 3.15. The symmetric decreasing rearrangement f^* of a function $f \in L^2(\Omega)$ is the spherically symmetric function whose level sets have equal volume to those of f ,

$$f^*(x) = \int_0^\infty \mathbb{1}_{U_t^*} dt.$$

Firstly, we are justified in calling f^* symmetric as

$$\frac{\partial}{\partial \theta} \int_0^\infty \mathbb{1}_{U_t^*} dt = \int_0^\infty \frac{\partial}{\partial \theta} \mathbb{1}_{U_t^*} dt = 0$$

since

$$\mathbb{1}_{U_t^*} = \begin{cases} 1 & |x| < |U_t|/\omega_N, \\ 0 & |x| \geq |U_t|/\omega_N. \end{cases}$$

Furthermore the level sets of f and f^* have the same measure, which is demonstrated in Figure 7. The code I wrote to produce these diagrams is included in appendix A. In the left plot we see a function $f(x)$ which is neither symmetric nor decreasing. Its symmetric decreasing rearrangement, f^* is shown in the right plot. Then we see that the level set U_t of f with $t = 2.16$, the horizontal blue line, has the same length as the level set of $t = 2.16$ of f^* , the horizontal green line. The level sets of f^* are the symmetric rearrangements of the level sets of f , which in 1 dimension are just symmetric intervals. This idea generalises to higher dimensions but instead of intervals we reconstruct into balls centred at the origin. It is important to note that when we refer to f^* as the symmetric *decreasing* rearrangement, f^* is decreasing in the sense that $f^*(x) = g(|x|)$ where g is a decreasing function. Finally, calling this a rearrangement is justified as we preserve the L^p norm.

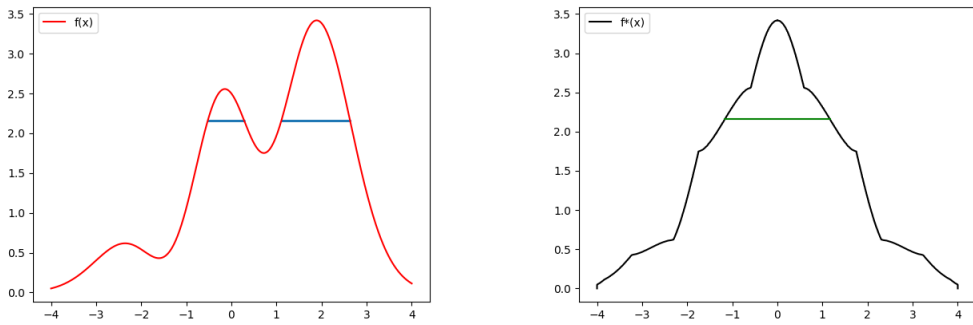


Figure 7: The left plot shows $f(x) = (x^2 + 3x + 5/2)(x^2 - 3x/2 + 1)e^{-x^2/2}$ and the right plot shows its symmetric decreasing rearrangement $f^*(x)$.

Lemma 3.18. For every $f \in L^p(\mathbb{R}^N)$, $1 \leq p < \infty$

$$\|f\|_p = \|f^*\|_p$$

Proof. Using Fubini's theorem in the second equality,

$$\begin{aligned} \int_{\mathbb{R}^N} |f(x)|^p dx &= \int_{\mathbb{R}^N} \int_0^\infty \mathbb{1}_{\{f(x)^p > t\}} dt dx = \int_0^\infty \mu(\{f(x)^p > t\}) dt \\ &= \int_0^\infty \mu(\{f(x) > s\}) p s^{p-1} ds = \int_0^\infty \mu(\{f^*(x) > s\}) p s^{p-1} ds \end{aligned}$$

from which we can retrace our steps with f replaced by f^* . □

The following result will prove useful.

Lemma 3.19 (The co-area formula). *The co-area formula states that for a non-negative, $L^2(\Omega)$ function f ,*

$$\int_{\Omega} g(x) |\nabla f(x)| dx = \int_0^\infty \int_{f^{-1}(t)} g(x) d\sigma dt.$$

Here $f^{-1}(t) = \{x \in \Omega : f(x) = t\}$ is the $N - 1$ dimensional level surface.

This is the generalisation of the idea that integrating over all of $\mathbb{R}^N$ is equivalent to an iterated integral over the spherical shells which are the level sets of the radius:

$$\int_{\mathbb{R}^N} g(x) dx = \int_0^\infty \left(\int_{\partial B_r(0)} g d\sigma \right) dr.$$

In fact this is the co-area formula where $f(x) = |x|$ [8].

4 Properties of Eigenfunctions and Eigenvalues

With the necessary preliminaries in place we can now say more about the Robin eigenfunctions and eigenvalues. We start by multiplying the eigenvalue equation (1.6) by the eigenfunction u and integrating over Ω .

$$-\int_{\Omega} u \nabla^2 u \, dx = \lambda^{\alpha} \int_{\Omega} u^2 \, dx.$$

Integration by parts yields:

$$-\int_{\partial\Omega} u \nabla u \cdot \boldsymbol{\nu} \, d\sigma + \int_{\Omega} \nabla u \cdot \nabla u \, dx = \lambda^{\alpha} \int_{\Omega} u^2 \, dx.$$

The Robin boundary condition states that on $\partial\Omega$, $\nabla u \cdot \boldsymbol{\nu} = -\alpha u$, hence

$$\alpha \int_{\partial\Omega} u^2 \, d\sigma + \int_{\Omega} |\nabla u|^2 \, dx = \lambda^{\alpha} \int_{\Omega} u^2 \, dx.$$

Assuming $u \not\equiv 0$ then the integral on the right-hand side is positive, meaning

$$\lambda^{\alpha}(\Omega) = \frac{\int_{\Omega} |\nabla u|^2 \, dx + \alpha \int_{\partial\Omega} u^2 \, d\sigma}{\int_{\Omega} u^2 \, dx}. \quad (4.1)$$

4.1 Variational Characterisation

In the following section I have adapted the proofs given for the Dirichlet case in Strauss [38] to the Robin case. The quotient expression for $\lambda^{\alpha}(\Omega)$ in (4.1) motivates the following definition:

Definition 4.1. The Rayleigh Quotient $R : H^1(\Omega) \rightarrow \mathbb{R}$ is given by:

$$R[u] = \frac{B[u, u]}{(u, u)_{L^2(\Omega)}} = \frac{\int_{\Omega} |\nabla u|^2 \, dx + \alpha \int_{\partial\Omega} u^2 \, d\sigma}{\int_{\Omega} u^2 \, dx}. \quad (4.2)$$

Euler, in his seminal work on the calculus of variations, remarked that

“Since the structure of the universe was made most perfect as designed by the wisest Creator, nothing in the world will occur in which no maximum or minimum rule is shining forth.” [33]

The following theorem casts the eigenvalue problem in this light.

Theorem 4.1. 1. The smallest eigenvalue of the Robin Laplacian on Ω , $\lambda_1^{\alpha}(\Omega)$, is the minimum of the Rayleigh quotient over $H^1(\Omega)$. That is,

$$\lambda_1^{\alpha}(\Omega) = \min_{u \in H^1(\Omega)} R[u] = \min_{u \in H^1(\Omega)} \frac{\int_{\Omega} |\nabla u|^2 \, dx + \alpha \int_{\partial\Omega} u^2 \, d\sigma}{\int_{\Omega} u^2 \, dx}.$$

2. The function which minimises the Rayleigh Quotient is an eigenfunction of the Robin Laplacian with corresponding eigenvalue $\lambda_1^{\alpha}(\Omega)$.

To prove this the following result will be especially useful, which follows from the first vanishing theorem in Appendix A.1 of Strauss by taking $\varphi \equiv 1$ [38].

Lemma 4.2 (The Fundamental Lemma of the Calculus of Variations). Suppose

$$\int_{\Omega} f \varphi \, dx = 0$$

for all $\varphi \in H^1(\Omega)$. Then $f = 0$ a.e.

Proof of Theorem 4.1. Suppose that the minimum is attained by some function u and we denote the corresponding Rayleigh quotient $R[u] = m^*$. We want to show that $-\nabla^2 u = m^* u$. To do this, we perturb the Rayleigh quotient's argument by some small trial function $\varepsilon v \in H^1(\Omega)$ and use the fact that

$$\left. \frac{d}{d\varepsilon} R[u + \varepsilon v] \right|_{\varepsilon=0} = 0.$$

When we differentiate with respect to ε and set it equal to zero we find that the following vanishes:

$$\begin{aligned} & \frac{(\int_{\Omega} \nabla u \cdot \nabla v \, dx) (\int_{\Omega} u^2 \, dx) - (\int_{\Omega} uv \, dx) (\int_{\Omega} |\nabla u|^2 \, dx)}{(\int_{\Omega} u^2 \, dx)^2} \\ & + \alpha \frac{(\int_{\Omega} u^2 \, dx) (\int_{\partial\Omega} uv \, d\sigma) - (\int_{\partial\Omega} u^2 \, d\sigma) (\int_{\Omega} uv \, dx)}{(\int_{\Omega} u^2 \, dx)^2} = 0. \end{aligned}$$

Formidable as this looks, it can be rendered harmless:

$$\begin{aligned} \int_{\Omega} \nabla u \cdot \nabla v \, dx + \alpha \int_{\partial\Omega} uv \, d\sigma &= \left(\frac{\int_{\Omega} |\nabla u|^2 \, dx + \alpha \int_{\partial\Omega} u^2 \, d\sigma}{\int_{\Omega} u^2 \, dx} \right) \int_{\Omega} uv \, dx \\ &= m^* \int_{\Omega} uv \, dx. \end{aligned}$$

Integrating the leftmost term by parts gives:

$$\begin{aligned} & \left[- \int_{\Omega} v \nabla^2 u \, dx + \int_{\partial\Omega} v \frac{\partial u}{\partial \nu} \, d\sigma \right] + \alpha \int_{\partial\Omega} uv \, d\sigma = m^* \int_{\Omega} uv \, dx \\ & \Rightarrow \int_{\Omega} (\nabla^2 u + m^* u) v \, dx = \int_{\partial\Omega} \left(\frac{\partial u}{\partial \nu} + \alpha u \right) v \, d\sigma. \end{aligned} \tag{4.3}$$

This is true for all trial functions v but we can make a strategic choice: let v vanish on the boundary. Then the right-hand term in (4.3) vanishes and we can use Lemma 4.2 to conclude

$$-\nabla^2 u = m^* u$$

almost everywhere. Then the left-hand side of (4.3) vanishes for all v and relaxing our restriction on v we see the right-hand side of (4.3) vanishes for all trial functions v so again by Lemma 4.2 we see that u satisfies the Robin boundary condition as well.

Lastly we have to prove that $R[u] = m^*$ is the smallest such eigenvalue. Let (u_j, λ_j^α) be any eigenpair of the Robin Laplacian. Then by the minimality of u ,

$$\begin{aligned} m^* &\leq \frac{\int_{\Omega} |\nabla u_j|^2 \, dx + \alpha \int_{\partial\Omega} u_j^2 \, d\sigma}{\int_{\Omega} u_j^2 \, dx} = \frac{\int_{\Omega} -u_j \nabla^2 u_j \, dx + \int_{\partial\Omega} u_j \frac{\partial u_j}{\partial \nu} \, d\sigma + \alpha \int_{\partial\Omega} u_j^2 \, d\sigma}{\int_{\Omega} u_j^2 \, dx} \\ &= \frac{\int_{\Omega} -u_j \nabla^2 u_j \, dx}{\int_{\Omega} u_j^2 \, dx} = \lambda_j^\alpha. \end{aligned} \tag{4.4}$$

The proof rests on the assumption that the minimum is indeed attained: how reasonable is this? We proved that any minimiser is an eigenfunction but we can run the proof in reverse and conclude that any eigenfunction is a minimiser, which we know exists due to the Lax-Milgram theorem. $\square$

In the previous proof we can see that the Rayleigh quotient of *any* eigefunction is its eigenvalue so we'd like to extend Theorem 4.1 to higher eigenvalues.

Theorem 4.3. 1. *Subsequent eigenvalues are given by:*

$$\lambda_n^\alpha(\Omega) = \min \{ R[v] : 0 \neq v \in H^1(\Omega), (v, u_{n-1}) = \dots = (v, u_1) = 0 \},$$

where $u_1, u_2, \dots, u_{n-1}$ are the eigenfunctions corresponding to eigenvalues $\lambda_1^\alpha \leq \lambda_2^\alpha \leq \dots \leq \lambda_{n-1}^\alpha$ respectively.

2. The minimising function is an eigenfunction, u_n , corresponding to the n -th eigenvalue.

Proof. Let $u(x)$ be a minimiser of the Rayleigh quotient satisfying the orthogonality constraint and m^* be the Rayleigh quotient evaluated on u . Proceeding exactly as in the proof of Theorem 4.1 we conclude

$$\int_{\Omega} (\nabla^2 u + m^* u) v \, dx = 0, \quad (4.5)$$

for all trial functions $v \in H^1(\Omega)$ orthogonal to $u_1, u_2, \dots, u_{n-1}$. Although u_j for any $j = 1, \dots, n-1$ doesn't satisfy the same orthogonality condition as v (as it isn't orthogonal to itself), we see that it does still satisfy equation (4.5). Using the self-adjointness of the Robin Laplacian (Lemma 3.12) and orthogonality of u with u_j :

$$\begin{aligned} \int_{\Omega} (\nabla^2 u + m^* u) u_j \, dx &= (\nabla^2 u, u_j) + (m^* u, u_j) \\ &= (u, \nabla^2 u_j) + m^* (u, u_j) \\ &= (m^* - \lambda_j)(u, u_j) = 0. \end{aligned} \quad (4.6)$$

Now let $h(x)$ be a trial function. Then we define

$$\tilde{v}(x) := h(x) - \sum_{k=1}^{n-1} c_k u_k(x), \quad c_k = \frac{(h, u_k)}{(u_k, u_k)}.$$

Then

$$\begin{aligned} (\tilde{v}, u_k) &= \left(h - \sum_{j=1}^{n-1} c_j u_j, u_k \right) = (h, u_k) - \sum_{j=1}^{n-1} c_j (u_j, u_k) \\ &= (h, u_k) - \sum_{j=1}^{n-1} c_j \delta_{jk} (u_j, u_k) = (h, u_k) - c_k (u_k, u_k) = (h, u_k) - (h, u_k) \\ &= 0. \end{aligned} \quad (4.7)$$

Thus $\tilde{v}(x)$ is orthogonal to each eigenfunction u_j for $j = 1, \dots, n-1$ and so it satisfies equation (4.5):

$$\begin{aligned} 0 &= \int_{\Omega} (\nabla^2 u + m^* u) \tilde{v} \, dx \\ &= \int_{\Omega} (\nabla^2 u + m^* u) \left(h(x) - \sum_{k=1}^{n-1} c_k u_k(x) \right) dx \\ &= \int_{\Omega} (\nabla^2 u + m^* u) h(x) \, dx - \sum_{k=1}^{n-1} c_k \int_{\Omega} (\nabla^2 u + m^* u) u_k(x) \, dx \\ &= \int_{\Omega} (\nabla^2 u + m^* u) h(x) \, dx. \end{aligned}$$

The final equality relies on equation (4.6). Since $h(x)$ was an arbitrary trial function, Lemma 4.2 implies that $-\nabla^2 u = m^* u$.

It remains to show that $m^* = \lambda_n^\alpha$. We know that m^* is greater than λ_j^α for $j = 1, \dots, n-1$ as the minimum in (4.3) is taken over a smaller set than in Theorem (4.1) so necessarily must be greater. Now if λ_k^α is an eigenvalue corresponding to u_k which is orthogonal to all u_i for $i = 1, \dots, n-1$ then proceeding exactly as in equation (4.4) we conclude that $m^* \leq \lambda_k^\alpha$. $\square$

4.2 Monotonicity of the Eigenvalues

We can ask about monotonicity of the eigenvalues with respect to both the boundary parameter α and the domain Ω . We'll see that the former is a positive result and provide a counterexample for the latter.

Firstly, inspection of the variational characterisation

$$\lambda_1^\alpha(\Omega) = \min_{u \in H^1(\Omega)} R[u] = \min_{u \in H^1(\Omega)} \frac{\int_\Omega |\nabla u|^2 dx + \alpha \int_{\partial\Omega} u^2 d\sigma}{\int_\Omega u^2 dx}$$

answers the first of our questions.

Lemma 4.4. *The n -th Robin eigenvalue λ_n^α is a monotonically increasing function of α .*

This holds for higher eigenvalues as the space over which we minimise doesn't affect the relationship between λ^α and α .

When introducing the Robin boundary condition we stated that the Dirichlet and Neumann conditions are the limiting cases of the Robin condition as $\alpha \rightarrow \infty$ and $\alpha \rightarrow 0$ respectively. We would like to show that the eigenvalues respect this limiting interpretation. To do this we need a variational characterisation for the Dirichlet and Neumann eigenvalues. Denoting

$$R_0[v] := \frac{\int_\Omega |\nabla v|^2 dx}{\int_\Omega |v|^2 dx},$$

it is possible to prove in the same fashion as Theorem 4.1 that the Neumann eigenfunctions are minimisers of $R_0[v]$ over $H^1(0)$, while the Dirichlet eigenfunctions are minimisers of $R_0[v]$ over the smaller space $H_0^1(\Omega) := \{u \in H^1(\Omega) : u(x) = 0 \text{ for all } x \in \partial\Omega\}$. Specifically,

$$\begin{cases} \eta_1 = \min_{H^1(\Omega)} R_0[v], \\ \mu_1 = \min_{H_0^1(\Omega)} R_0[v]. \end{cases}$$

Lemma 4.5. *For $\alpha > 0$,*

$$\eta_1(\Omega) \leq \lambda_1^\alpha(\Omega) \leq \mu_1(\Omega).$$

Proof. For the first inequality

$$\int_\Omega |\nabla u|^2 dx \leq \int_\Omega |\nabla u|^2 dx + \alpha \int_{\partial\Omega} u^2 d\sigma.$$

For the second inequality,

$$\begin{aligned} \min_{H_0^1(\Omega)} \frac{\int_\Omega |\nabla u|^2 dx}{\int_\Omega u^2 dx} &= \min_{H_0^1(\Omega)} \frac{\int_\Omega |\nabla u|^2 dx + \alpha \int_{\partial\Omega} u^2 d\sigma}{\int_\Omega u^2 dx} \\ &\geq \min_{H^1(\Omega)} \frac{\int_\Omega |\nabla u|^2 dx + \alpha \int_{\partial\Omega} u^2 d\sigma}{\int_\Omega u^2 dx}, \end{aligned}$$

as the latter minimum is taken over a larger set. □

It is in fact possible to show that we recover the Neumann and Dirichlet eigenvalues in the respective limits $\alpha \rightarrow 0$ and $\alpha \rightarrow \infty$. The former is clear to see from the variational characterisation however the latter is more delicate and doesn't necessarily hold for $\alpha \rightarrow -\infty$ [30].

We are less fortunate when we turn our attention to the second question. We saw in the Robin problem on the interval that the first eigenvalue was a decreasing function of the interval half length. Unfortunately such statements do not hold for Robin eigenvalues in higher dimensions, while they do in the Dirichlet case.

Lemma 4.6. *The Dirichlet eigenvalues are monotone with respect to their domain. That is, if $\Omega_1 \subset \Omega_2$ such that the closure of Ω_1 is a strict subset of the interior of Ω_2 then*

$$\mu_1(\Omega_1) \geq \mu_1(\Omega_2).$$

To prove this we need the following extension result from Evans and Gariepy [9].

Theorem 4.7 (Extension Theorem). *Suppose $\Omega_1 \subset \Omega_2$ such that the closure of Ω_1 is a strict subset of the interior of Ω_2 . There exists a bounded linear operator $E : H^1(\Omega_1) \rightarrow H^1(\mathbb{R}^N)$ such that*

$$Ef = f$$

on Ω_1 and

$$\text{sppt}(Ef) = \{x \in \mathbb{R}^N : Ef(x) = 0\} \subset \Omega_2$$

for all $f \in H^1(\Omega_1)$. We call Ef the extension of F .

In essence we cannot simply set Ef to be zero on $\partial\Omega_1$ as the discontinuity this introduces might be so sharp that Ef is no longer weakly differentiable. We need the distance between $\partial\Omega_1$ and $\partial\Omega_2$ to be positive in order to bring the value of f to zero inside Ω_2 so that $Ef \in H_0^1(\Omega_2)$.

Proof. We would like to show that $H_0^1(\Omega_1) \subset H_0^1(\Omega_2)$. By the extension theorem, any function $f \in H_0^1(\Omega_1)$ can be extended to $Ef \in H^1(\Omega_2)$ which has support contained in Ω_2 meaning $Ef \in H_0^1(\Omega_2)$. Thus we can view any function in $H_0^1(\Omega_1)$ as the restriction of a function in $H_0^1(\Omega_2)$, establishing the inclusion. Then

$$\mu_1(\Omega_1) = \min_{v \in H_0^1(\Omega_1)} \frac{\int_{\Omega} |\nabla^2 v|^2 dx}{\int_{\Omega} |v|^2 dx} \geq \min_{v \in H_0^1(\Omega_2)} \frac{\int_{\Omega} |\nabla^2 v|^2 dx}{\int_{\Omega} |v|^2 dx} = \mu_1(\Omega_2),$$

where once again the latter minimum is smaller as it is taken over a larger set. $\square$

Before we provide a counter-example, there are specific instances in which we can expect monotonicity of the Robin eigenvalues.

Definition 4.2. A homothety of a set Ω by a constant $t > 0$ is the set

$$t\Omega := \{tx : x \in \Omega\}$$

Using the variational characterisations of the eigenvalues we see that the Dirichlet and Neumann eigenvalues obey:

$$\begin{cases} \mu_n(t\Omega) = t^{-2}\mu_n(\Omega), \\ \eta_n(t\Omega) = t^{-2}\eta_n(\Omega). \end{cases}$$

Whereas letting $s = \frac{1}{t}x$ and noticing that $dx = t^N ds$, $d\sigma = t^{N-1}d\tilde{\sigma}$ and $\nabla_x = \frac{1}{t}\nabla_s$, we find

$$\frac{\int_{t\Omega} |\nabla u|^2 dx + t\alpha \int_{\partial t\Omega} u^2 d\sigma}{\int_{t\Omega} u^2 dx} = \frac{\int_{\Omega} \frac{1}{t^2} |\nabla u|^2 t^N ds + t\alpha \int_{\partial\Omega} u^2 t^{N-1} d\tilde{\sigma}}{\int_{\Omega} u^2 t^N dx}.$$

Then taking the infimum over the appropriate space we find

$$\lambda_n^{t\alpha}(t\Omega) = \frac{1}{t^2} \lambda_n^\alpha(\Omega),$$

where now the boundary parameter comes into play. This lack of clean scaling, along with the absence of domain monotonicity, is one of the reasons that the Robin problem does not lend itself to the same treatment as the Dirichlet problem. Although we don't have monotonicity for general domains, we do for homothetic domains [15]:

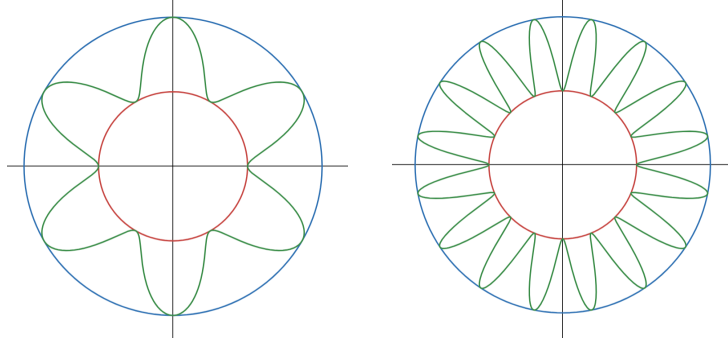


Figure 8: Domains Ω_6 and Ω_{16}

Lemma 4.8. *Fix $\alpha > 0$. Then Robin eigenvalues are monotone with respect to homothetic domains. That is, for $t \geq 1$,*

$$\lambda_n^\alpha(t\Omega) \leq \lambda_n^\alpha(\Omega).$$

Proof.

$$\begin{aligned} \lambda_n^\alpha(t\Omega) &= \frac{1}{t^2} \lambda_n^{\alpha/t}(\Omega) = \min_{u \in H^1(\Omega)} \frac{\frac{1}{t^2} \int_{\Omega} |\nabla u|^2 dx + \frac{\alpha}{t^3} \int_{\partial\Omega} u^2 d\sigma}{\int_{\Omega} u^2 dx} \\ &\leq \min_{u \in H^1(\Omega)} \frac{\int_{\Omega} |\nabla u|^2 dx + \alpha \int_{\partial\Omega} u^2 d\sigma}{\int_{\Omega} u^2 dx}. \end{aligned}$$

□

Giorgi and Smits provided a counter-example to general domain monotonicity for Robin eigenvalues as follows [12]. First fix $\alpha < 0$. Then by taking u to be constant in the Rayleigh quotient we see that

$$\lambda_1^\alpha(\Omega) \leq \alpha \frac{|\partial\Omega|}{|\Omega|}.$$

We can construct a sequence of domains Ω_n sandwiched between $B_1(0)$ and $B_2(0)$ such that $|\partial\Omega_n| \rightarrow \infty$ as $n \rightarrow \infty$, for example those shown in Figure 8. Then for n sufficiently large we have

$$\lambda_1^\alpha(\Omega_n) < \lambda_1^\alpha(B_1(0)) < \lambda_1^\alpha(B_2(0)) < 0,$$

ruling out any monotonicity for $\alpha < 0$. Now fix $\alpha > 0$. Due to the scaling properties of eigenvalues, there exists an $\varepsilon > 0$ such that

$$0 < \lambda_1^\alpha(B_2(0)) < \lambda_1^\alpha(B_1(0)) < \mu_1(B_{1+\varepsilon}(0)).$$

Then by *Theorem 4.3* of Daners and Dancer [6], there exists a sequence of domains Ω_n sandwiched between $B_1(0)$ and $B_{1+\varepsilon}(0)$ with $|\partial\Omega_n| \rightarrow \infty$ such that

$$\lim_{n \rightarrow \infty} \lambda_1^\alpha(\Omega_n) = \mu_1(B_{1+\varepsilon}(0)).$$

Then for a suitably large n ,

$$0 < \lambda_1^\alpha(B_2(0)) < \lambda_1^\alpha(B_1(0)) < \lambda_1^\alpha(\Omega_n).$$

Thus for no α can we expect domain monotonicity.

4.3 Nodal Geometry

Recalling that the solutions to the wave equation are Laplacian eigenfunctions multiplied by a trigonometric function, when we let a membrane vibrate at a pure frequency we see that the region on which the eigenfunctions vanish persists in time. Physically, if we spread salt on a plate of shape Ω and vibrate it at a certain frequency, the salt will collect in stationary patterns. This type of experiment was first conducted by Chladni in the late 19th century and so such plates are called Chladni plates [26]. Some examples are shown in Figure 9. Equilibrium points where the

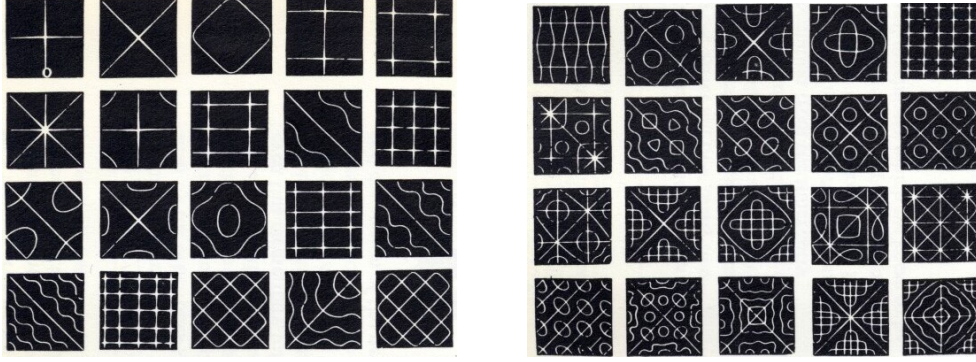


Figure 9: Nodal sets on Chladni plates [40]

salt collects have a name.

Definition 4.3. We define for an eigenfunction u_k on Ω the nodal set given by

$$\mathcal{N} = \{x \in \Omega : u_k(x) = 0\}.$$

We define a connected component of $\Omega \setminus \mathcal{N}$ to be a nodal domain on which the eigenfunction has a sign. On the interval $(0, L)$ the n -th Dirichlet eigenfunction is $\sin(\frac{n\pi x}{L})$. The set of nodal points $\{\frac{kL}{n} : k = 0, \dots, n\}$ contains $n + 1$ points where a guitar player could place a finger and not interrupt the vibration of the string. The n -th eigenfunction divides the interval into n nodal domains; is this always the case?

Theorem 4.9 (Courant's Nodal Domain Theorem, 1923). *The n -th eigenfunction divides the domain Ω into at most n nodal domains.*

We adapt the proof given in Levitin, Mangoubi and Polterovich to the Robin case [26].

Proof. Suppose u is a Robin eigenfunction with eigenvalue λ^α and that u divides Ω into a disjoint union of at least k nodal domains, $\Omega_1, \dots, \Omega_k, \Omega_{k+1}, \dots$. We aim to show that $\lambda^\alpha > \lambda_k^\alpha$. First define the restriction of u to each of the nodal domains by

$$\psi_i(x) := \begin{cases} u(x) & x \in \Omega_i, \\ 0 & \text{otherwise.} \end{cases}$$

It can be shown that $\psi_i \in H^1(\Omega_i)$. Define $\Psi = \text{span}\{\psi_1, \dots, \psi_k\}$ and let

$$f := \sum_{i=1}^k \beta_i \psi_i \in \Psi.$$

Since $\dim(\Psi) = k$ we can choose the coefficients β_i in such a way that f is orthogonal to the first $k - 1$ eigenfunctions. These $k - 1$ restraints leave one degree of freedom to normalise f . Then with computing the Rayleigh quotient of f in mind,

$$\begin{aligned}
\int_{\Omega} |\nabla f|^2 dx &= \sum_{i=1}^k \int_{\Omega_i} |\beta_i \nabla \psi_i|^2 dx = \sum_{i=1}^k \beta_i^2 \int_{\Omega_i} \nabla \psi_i \cdot \nabla \psi_i dx \\
&= \sum_{i=1}^k \beta_i^2 \left[- \int_{\Omega_i} \psi_i \nabla^2 \psi_i dx + \int_{\partial \Omega_i} \psi_i \frac{\partial \psi_i}{\partial \nu} d\sigma \right] \\
&= \sum_{i=1}^k \beta_i^2 \left[\int_{\Omega_i} \lambda^\alpha \psi_i^2 dx + \int_{\partial \Omega_i \cap \partial \Omega} \psi_i \frac{\partial \psi_i}{\partial \nu} d\sigma + \int_{\partial \Omega_i \cap \Omega} \psi_i \frac{\partial \psi_i}{\partial \nu} d\sigma \right] \\
&= \sum_{i=1}^k \left[\int_{\Omega_i} \lambda^\alpha (\beta_i \psi_i)^2 dx - \alpha \int_{\partial \Omega_i \cap \partial \Omega} (\beta_i \psi_i)^2 d\sigma \right] \\
&= \lambda^\alpha \int_{\Omega} f^2 dx - \alpha \int_{\partial \Omega} f^2 d\sigma.
\end{aligned}$$

We've used the fact that ψ_i vanishes on the internal component of $\partial \Omega_i$ and obeys the Robin boundary condition on the external component. If we evaluate the Rayleigh quotient at f we find that $R[f]$ must be λ^α . Assume that $\lambda^\alpha = \lambda_k^\alpha$. As f is a function orthogonal to the first $k-1$ eigenvalues, the Rayleigh quotient of which is λ_k^α , by Theorem 4.3 f is an eigenfunction with eigenvalue λ_k^α . By Theorem 3.17, this means that f is real analytic on the whole of Ω . But $f|_{\Omega_{k+1}} = 0 \Rightarrow f \equiv 0$ on all of Ω . Therefore an eigenfunction corresponding to λ_k^α has at most k nodal domains. $\square$

Courant's theorem is powerful as it gives us lots of information about the first two eigenfunctions, summarised in the following corollaries.

Corollary 4.9.1. *The first eigenfunction can be chosen to be positive everywhere on Ω .*

Corollary 4.9.2. *The first eigenvalue is simple.*

Proof. Suppose that λ_1 has two associated eigenfunctions, u and v . Then they can be chosen to be orthogonal. But by Corollary 4.9.1 both are positive so their L^2 inner-product must be positive, contradicting their orthogonality. $\square$

Corollary 4.9.3. *The second eigenfunction has precisely two nodal domains.*

Proof. By Courant's theorem, u_2 has at most two nodal domains. By Lemma 3.6, we know that u_1 and u_2 are orthogonal. As u_1 is positive, u_2 must change sign, meaning it has at least two nodal domains. $\square$

Corollary 4.9.4. *The first eigenfunction is the only eigenfunction which does not change sign.*

4.4 The Minimum Principle

We now collect a minimum principle for the eigenfunctions, following Evans [8]. Recall that a general elliptic operator has the form

$$\mathcal{L}u = - \sum_{i,j} a_{i,j} \partial_{x_i} \partial_{x_j} u + \sum_i b_i \partial_{x_i} u + cu,$$

subject to the ellipticity constraint on a_{ij} , equation (3.5). In this setting eigenfunctions are solutions of $\mathcal{L}u = 0$ where $a_{i,j} = \delta_{i,j}$, $b_i = 0$ and $c = \lambda$.

Theorem 4.10 (Weak and Strong Minimum Principles).

1. Assume that $c \geq 0$ and that $u \in C^2(\Omega) \cap C(\overline{\Omega})$. If $\mathcal{L}u = 0$ in Ω then

$$\min_{\overline{\Omega}} |u| = \min_{\partial \Omega} |u|.$$

2. Assume further that Ω is connected. Then provided $Lu \geq 0$, u attaining a nonpositive minimum over $\overline{\Omega}$ at an interior point implies that u is constant.

The condition $Lu \geq 0$ implies that u is a supersolution and the first and second statements are referred to as weak and strong minimum principles respectively. This is because the latter implies the former but the former implying the latter relies on the connectedness of Ω . It is not difficult to imagine a function which is piecewise constant on a disconnected domain and so attains a minimum at an interior point.

Corollary 4.10.1. *If u is a non-constant eigenfunction with positive eigenvalue then u is a continuous function and the set $\overline{\Omega}$ is compact so u attains a minimum. By part 1 this minimum is attained on the boundary and by part 2 it is positive.*

5 Shape Optimisation For the Robin Problem

It was first conjectured by Lord Rayleigh in 1877 in his treatise *The Theory of Sound* [36] that

“If the area of a membrane be given, there must evidently be some form of boundary for which the pitch (of the principal tone) is the gravest possible, and this form can be no other than the circle.”

Rayleigh’s conjecture was proven by Faber in two dimensions in 1923 [10] and extended to arbitrary dimensions by Krahn in 1925 [20]. Krahn’s doctoral supervisor was Courant, who in turn was supervised by Hilbert.

Theorem 5.1 (Faber-Krahn Inequality, 1925). *Let $\mu_1(\Omega)$ be the first Dirichlet eigenvalue on the bounded domain $\Omega \subset \mathbb{R}^N$ and Ω^* be the ball of equal volume. Then*

$$\mu_1(\Omega) \geq \mu_1(\Omega^*) \quad (5.1)$$

with equality holding only for the ball.

An analogous result was later proved for the maximisation of the first non-trivial Neumann eigenvalue, first in two dimensions by Szegő in 1954 [39] and then in higher dimensions by Weinberger in 1956 [42]. As we’ve seen repeatedly, the Robin problem is not amenable to the same techniques as the Dirichlet problem and so it took until 1986 for the two-dimensional analogue to be proven by Bossel [3] and another two decades for the higher dimensional extension courtesy of Daners [7]. Both these results hold only for $\alpha > 0$.

5.1 The First Robin Eigenvalue

The proof given below is that of Daners [7]. I have tried to provide additional insight and to make the presentation clearer.

Theorem 5.2 (Bossel-Daners Inequality, 2006). *Let $\alpha > 0$ and Ω be a bounded, Lipschitz domain in $\mathbb{R}^N$. If Ω^* is the ball with $|\Omega^*| = |\Omega|$, then*

$$\lambda_1^\alpha(\Omega) \geq \lambda_1^\alpha(\Omega^*).$$

The proof will come in two acts. Firstly, we define a functional of the level sets of the first eigenfunction and establish a lower bound for the first eigenvalue on Ω . Then we will use a rearrangement argument to establish an upper bound for the first eigenvalue on the ball, then conclude by matching the bounds up.

Denote the eigenfunction corresponding to $\lambda_1^\alpha(\Omega)$ by u . Then we can choose for u to be positive inside Ω due to Corollary 4.9.1 and we are free to normalise so that $\|u\|_\infty = 1$. Denote

$$m = \min_{x \in \Omega} u(x).$$

Then by Theorem 4.10, $m > 0$ and is attained on $\partial\Omega$. Consider the level sets of u , denoted $U_t = \{x \in \Omega : u(x) > t\}$, which are open as u is continuous in Ω by Theorem 3.17. The boundary of each U_t has an internal component and an external one:

$$S_t := \{x \in \Omega : u(x) = t\}$$

is the level surface and so the interior boundary component. The external boundary component is $\Gamma_t := \partial U_t \cap \partial\Omega$. Furthermore $S_t = \emptyset$ for $t \notin (m, 1]$. For a positive function $\varphi \in C(\Omega)$ and a level set $U_t \subset \Omega$ with positive measure, we define the functional

$$H_\Omega[U_t, \varphi] := \frac{1}{|U_t|} \left(\int_{S_t} \varphi d\sigma + \alpha |\Gamma_t| - \int_{U_t} \varphi^2 dx \right).$$

Though a little contrived, we immediately see its use in the following lemma.

Lemma 5.3. *For all $t \in (m, 1)$ such that $|U_t| > 0$, we have*

$$\lambda_1^\alpha(\Omega) = H_\Omega \left[U_t, \frac{|\nabla u|}{u} \right].$$

Proof. Multiplying the equation $-\nabla^2 u = \lambda_1^\alpha u$ by $1/u$ and integrating over U_t gives:

$$\begin{aligned} \lambda_1^\alpha |U_t| &= \int_{U_t} \lambda_1^\alpha dx = - \int_{U_t} \frac{1}{u} \nabla^2 u dx = \int_{U_t} \nabla u \cdot \nabla \left(\frac{1}{u} \right) dx - \int_{\partial U_t} \frac{1}{u} \nabla u \cdot \boldsymbol{\nu} d\sigma \\ &= - \int_{U_t} \left(\frac{|\nabla u|}{u} \right)^2 dx - \int_{\Gamma_t} \frac{1}{u} \frac{\partial u}{\partial \boldsymbol{\nu}} d\sigma - \int_{S_t} \frac{1}{u} \frac{\partial u}{\partial \boldsymbol{\nu}} d\sigma. \end{aligned}$$

As Γ_t is the external boundary, the Robin condition implies that $\frac{1}{u} \frac{\partial u}{\partial \boldsymbol{\nu}} = -\alpha$. Using the fact that ∇u is normal to level surfaces of u , $\nabla u \cdot \boldsymbol{\nu} = -|\nabla u|$ on the internal boundary S_t . The reason for the sign is that $\boldsymbol{\nu}$ points out from the level surfaces whereas by definition u attains a minimum over $\overline{U_t}$ on S_t meaning ∇u points inwards. Thus

$$\lambda_1^\alpha |U_t| = - \int_{U_t} \left(\frac{|\nabla u|}{u} \right)^2 dx + \alpha |\Gamma_t| + \int_{S_t} \frac{|\nabla u|}{u} d\sigma,$$

finishing the proof. □

We can relate our functional H on any function to the first eigenvalue.

Theorem 5.4. *Let $\varphi \in C(\Omega)$ be non-negative and set*

$$w := \varphi - \frac{|\nabla u|}{u}.$$

Then for almost all $t \in (m, 1)$,

$$H_\Omega[U_t, \varphi] = \lambda_1^\alpha(\Omega) + \frac{1}{|U_t|} \left(\int_{S_t} w d\sigma - 2 \int_t^1 \frac{1}{\tau} \int_{S_\tau} w d\sigma d\tau - \int_{U_t} w^2 dx \right).$$

Proof. We begin by noting that

$$\varphi^2 = w^2 + 2 \frac{w}{u} |\nabla u| + \frac{|\nabla u|^2}{u^2}.$$

Then using the above expression,

$$\begin{aligned} H_\Omega[U_t, \varphi] &= \frac{1}{|U_t|} \left\{ \int_{S_t} \left(w + \frac{|\nabla u|}{u} \right) d\sigma + \alpha |\Gamma_t| \right. \\ &\quad \left. - \int_{U_t} \left(w^2 + 2 \frac{w}{u} |\nabla u| + \frac{|\nabla u|^2}{u^2} \right) dx \right\}. \end{aligned}$$

We collect the three terms independent of w and use Lemma 5.3 to conclude

$$H_\Omega[U_t, \varphi] = \lambda_1^\alpha(\Omega) + \frac{1}{|U_t|} \left(\int_{S_t} w d\sigma - \int_{U_t} w^2 dx - 2 \int_{U_t} \frac{w}{u} |\nabla u| dx \right).$$

Then the lattermost term can be treated with the co-area formula, Lemma 3.19, as $u^{-1}(\tau) = S_\tau$:

$$\int_{U_t} \frac{w}{u} |\nabla u| dx = \int_t^1 \int_{S_\tau} \frac{w}{u} d\sigma d\tau = \int_t^1 \frac{1}{\tau} \int_{S_\tau} w d\sigma d\tau, \quad (5.2)$$

using the fact that $u = \tau$ on S_τ . Recombining gives the desired result. □

Denote the set of functions which are less than the boundary parameter α in some small neighbourhood of the boundary by

$$M_\alpha := \left\{ f \in C(\Omega) : \limsup_{x \rightarrow z} f(x) \leq \alpha \ \forall z \in \partial\Omega \right\}.$$

Let us establish a lower bound for $\lambda_1^\alpha(\Omega)$. Then later we hope to show that the same bound lies above $\lambda_1^\alpha(B)$.

Theorem 5.5. *With the above notation, for all $\varphi \in M_\alpha$ there exists $t \in (m, 1)$ such that*

$$\lambda_1^\alpha(\Omega) \geq H_\Omega[U_t, \varphi].$$

Corollary 5.5.1.

$$\lambda_1^\alpha(\Omega) \geq \inf_{t \in (m, 1)} H_\Omega[U_t, \varphi]$$

for all $\varphi \in M_\alpha$.

To prove this we will need a few ingredients. First we need a relationship between the measure of the level surfaces and the surface measure of $\partial\Omega$.

Lemma 5.6. *There exist $c > 0$ and $t_1 \in (m, 1)$ such that $|S_t| \leq c |\partial\Omega|$ for all $t \in (m, t_1]$.*

The proof of this involves approximating the surface S_t by open cubes and establishing an inequality between the surface measure of each cube and the surface measure of $\partial\Omega$ using the inverse function theorem. Then using the compactness of S_t , we conclude that the cover must have a finite subcover and so we can sum over finitely many cubes to bound the surface measure of S_t .

Next we treat the term in equation (5.2). We will need the definition of absolute continuity, a stronger property than uniform continuity [37].

Definition 5.1. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is absolutely continuous if for all $\varepsilon > 0$ there exists $\delta > 0$ such that for any finite collection of non-empty, pairwise disjoint intervals $\{(x_k, y_k)\}_{k=1}^m$ with total length less than δ , that is

$$\sum_{k=1}^m |y_k - x_k| < \delta,$$

the total difference between the function values at the endpoints is less than ε , that is

$$\sum_{k=1}^m |f(y_k) - f(x_k)| < \varepsilon.$$

Then thanks to *Theorem 8.18* of [37], we can reformulate this more easily as $f : [a, b] \rightarrow \mathbb{R}$ is absolutely continuous if and only if $f(x) = f(a) + \int_a^x f'(t) dt$ and $f' \in L^1((a, b))$.

Lemma 5.7. *Suppose $\varphi \in C(\Omega) \cap L_{loc}^1(\Omega)$ is non-negative. Let w be as in Lemma 5.4 and define*

$$F(t) := \int_t^1 \frac{1}{\tau} \int_{S_\tau} w d\sigma d\tau$$

for $t \in (m, 1)$. Then F is absolutely continuous on $[\varepsilon, 1)$ for all $\varepsilon \in (0, 1)$ and by the Fundamental Theorem of Calculus,

$$\frac{d}{dt} F(t) = -\frac{1}{t} \int_{S_t} w d\sigma,$$

for almost all $t \in (0, 1)$.

Proof. As

$$F(t) = \int_t^1 f(\tau) d\tau$$

where

$$f(\tau) := \frac{1}{\tau} \int_{S_\tau} w d\sigma,$$

F is absolutely continuous provided f is integrable on $(0, 1)$. Fix $\varepsilon > 0$.

$$f(\tau) = \frac{1}{\tau} \int_{S_\tau} w d\sigma = \frac{1}{\tau} \int_{S_\tau} \varphi d\sigma - \frac{1}{\tau} \int_{S_\tau} \frac{|\nabla u|}{u} d\sigma. \quad (5.3)$$

We now show that both terms on the right-hand side are integrable on $(\varepsilon, 1)$. The function φ is locally integrable by hypothesis and $\frac{|\nabla u|}{u}$ is bounded on U_ε as $u \in (\varepsilon, 1]$ and $|\nabla u|$ is a continuous function on a compact set. By the co-area formula,

$$\int_\varepsilon^1 \frac{1}{\tau} \int_{S_\tau} \varphi d\sigma d\tau = \int_{U_\varepsilon} \frac{\varphi}{u} |\nabla u| dx < \infty.$$

Doing the same for the second term in equation (5.3), we obtain

$$\int_\varepsilon^1 \frac{1}{\tau} \int_{S_\tau} \frac{|\nabla u|}{u} d\sigma d\tau = \int_{U_\varepsilon} \frac{|\nabla u|^2}{u^2} dx < \infty.$$

It follows that $f(\tau) \in L^1((0, 1))$ and so $F(t)$ is absolutely continuous on $[\varepsilon, 1)$ and differentiable almost everywhere with derivative

$$F'(t) = -f(t) = -\frac{1}{t} \int_{S_t} w dx.$$

As $\varepsilon > 0$ was arbitrary the claim holds for almost all $t \in (0, 1)$. $\square$

Lastly we will need to be able to bound w close to the boundary.

Lemma 5.8. *Let $\varphi \in M_\alpha$. Then for all $\varepsilon > 0$ there exists $\delta > 0$ such that when $d(x, \partial\Omega) < \delta$,*

$$\varphi(x) \leq \varepsilon + \frac{|\nabla u|}{u}.$$

Phrased differently, $w(x) \leq \varepsilon$.

Proof. Let $\varphi \in M_\alpha$ and let $\varepsilon > 0$. Then $\frac{|\nabla u|}{u}$ is continuous on Ω and so is uniformly continuous on $\bar{\Omega}$ which is compact so there exists $\delta_0 > 0$ such that

$$\left| \frac{|\nabla u(x)|}{u(x)} - \frac{|\nabla u(z)|}{u(z)} \right| < \frac{\varepsilon}{2} \quad (5.4)$$

for all $x, z \in \bar{\Omega}$ with $|x - z| < \delta_0$. Since $\varphi \in M_\alpha$, $\limsup_{x \rightarrow z} \varphi(x) \leq \alpha$. Therefore there exists $r > 0$ such that $\sup\{\varphi(x) : x \in \Omega, |x - z| < r\} \leq \alpha + \frac{\varepsilon}{2}$ for all $z \in \partial\Omega$. So

$$\varphi(x) - \alpha \leq \frac{\varepsilon}{2} \quad (5.5)$$

for all $x \in \Omega$ within r of the boundary. Let $\delta := \min\{\delta_0, r\}$. By the boundary condition and the Cauchy-Schwarz inequality:

$$-\alpha u = \nabla u \cdot \boldsymbol{\nu} \Rightarrow \alpha u = |\nabla u \cdot \boldsymbol{\nu}| \leq |\nabla u| \Rightarrow \alpha \leq \frac{|\nabla u|}{u}. \quad (5.6)$$

Putting it all together we see

$$\begin{aligned}
w(x) &= \varphi(x) - \frac{|\nabla u(x)|}{u(x)} = \varphi(x) - \alpha + \alpha - \frac{|\nabla u(x)|}{u(x)} \\
&\stackrel{(I)}{\leq} \frac{\varepsilon}{2} + \alpha - \frac{|\nabla u(x)|}{u(x)} \\
&\stackrel{(II)}{\leq} \frac{\varepsilon}{2} + \frac{|\nabla u(z)|}{u(z)} - \frac{|\nabla u(x)|}{u(x)} \\
&\stackrel{(III)}{\leq} \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.
\end{aligned}$$

Where inequality (I) follows from equation (5.5) as x is within r of $\partial\Omega$, (II) follows from equation (5.6) for any $z \in \partial\Omega$ and (III) follows from equation (5.4) as $d(x, \partial\Omega) < \delta_0$. $\square$

Armed with Lemmas 5.7 and 5.8, we are equipped to prove Theorem 5.5.

Proof of Theorem 5.5. Suppose for contradiction that there exists $\varphi \in M_\alpha$ such that $\lambda_1^\alpha(\Omega) < H_\Omega[U_t, \varphi]$ for all $t \in (m, 1)$. Then with w and F as above, by Theorem 5.4

$$\begin{aligned}
\lambda_1^\alpha(\Omega) < H_\Omega[U_t, \varphi] &= \lambda_1^\alpha(\Omega) + \frac{1}{|U_t|} \left(\int_{S_t} w \, d\sigma - 2F(t) - \int_{U_t} w^2 \, dx \right) \\
&\Rightarrow 2F(t) < \int_{S_t} w \, d\sigma - \int_{U_t} w^2 \, dx
\end{aligned}$$

for almost all $t \in (m, 1)$. By Lemma 5.7,

$$\frac{d}{dt} (t^2 F(t)) = t \left(- \int_{S_t} w \, d\sigma + 2F(t) \right) < -t \int_{U_t} w^2 \, dx \leq 0,$$

for almost all $t \in (m, 1)$ so the function $t^2 F(t)$ is strictly decreasing on $(m, 1)$. As $F(1) = 0$ and F is continuous there exist $l > 0$ and $t_0 \in (m, 1)$ such that $F(t) > l$ for all $t \in (m, t_0]$. We show that this contradicts the assumption $\varphi \in M_\alpha$. By Lemma 5.6 there exist $t_1 \in (m, t_0]$ and $c > 0$ such that $|S_t| \leq c |\partial\Omega|$ for all $t \in (m, t_1]$. Letting $\varepsilon = \frac{l}{c|\partial\Omega|}$, by Lemma 5.8 there exists a $\delta > 0$ such that $w(x) \leq \varepsilon$ for all $x \in \Omega$ within δ of $\partial\Omega$. Since u attains a strict minimum on $\partial\Omega$ there exists $t \in (m, t_1]$ such that $d(x, \partial\Omega) < \delta$ for all $x \in S_t$. In essence, we can choose a t small enough that the level surface S_t is contained entirely within the δ -strip attached to the boundary. Then

$$0 < 2l < 2F(t) < \int_{S_t} w \, d\sigma \leq \int_{S_t} \varepsilon \, d\sigma = \varepsilon |S_t| \leq \varepsilon c |\partial\Omega| = l,$$

but $0 < 2l < l$ is clearly a contradiction, completing the proof. $\square$

Now begins the second act. From now on all balls will be centred at the origin by translational symmetry of the Laplacian and Ω^* is the symmetric rearrangement of Ω . The first eigenfunction and eigenvalue on Ω^* are denoted u^* and $\lambda_1^\alpha(\Omega^*)$ respectively. As before we can normalise so that $u^* > 0$ and $\|u^*\|_\infty = 1$. As u^* is radially symmetric it is constant on $\partial\Omega^*$. This means that on the boundary

$$\frac{\partial u^*}{\partial \nu} = -|\nabla u^*| = -\alpha u^* \Rightarrow \frac{|\nabla u^*|}{u^*} \in M_\alpha. \quad (5.7)$$

We've used once again the argument that u^* obtains a minimum on the boundary and so ∇u^* and ν point in opposite directions. $\frac{|\nabla u^*|}{u^*}$ inherits radial symmetry from u^* and so we define the radial function:

$$g(|x|) := \frac{|\nabla u^*(x)|}{u^*(x)}.$$

Lemma 5.9. *The function defined above,*

$$g : (0, R) \rightarrow (0, \infty),$$

where R is the radius of Ω^ , is strictly increasing.*

Referring back to Section 2.5, eigenfunctions on the ball are given by Bessel functions. An encyclopedic account of properties of Bessel functions is given by Watson in [41], including the property stated above.

We need a way to communicate between functions on Ω^* and functions on Ω . Let $r : [m, 1] \rightarrow [0, R]$ be the function such that $U_t^* = B_{r(t)}$ is the symmetric rearrangement of U_t . Explicitly $\omega_N r(t)^N = |U_t|$. With this definition, $r(m) = R$ and $r(1) = 0$.

We construct a rearrangement of $|\nabla u^*(x)|/u^*(x)$ that lives in M_α . Define

$$\begin{aligned} \phi : \Omega &\rightarrow (0, \infty) \\ : x &\mapsto g(r(u(x))). \end{aligned}$$

Let's unpack this definition. $u : \Omega \rightarrow (m, 1]$ is the eigenfunction on Ω which maps $x \in \Omega$ to some $t \in (m, 1]$, meaning $x \in S_t$ which is the boundary of U_t . The symmetric rearrangement of U_t has radius $r(t)$ and the function $|\nabla u^*|/u^*$ evaluated anywhere on U_t^* has value $g(r(t))$.

Lemma 5.10. *The function ϕ belongs to M_α . Furthermore,*

$$\lambda_1^\alpha(\Omega^*) = H_{\Omega^*} \left[U_t^*, \frac{|\nabla u^*|}{u^*} \right] \leq H_\Omega[U_t, \phi]$$

for all $t \in (m, 1)$.

Proof. Since g is increasing (by Lemma 5.9), $\{x \in \Omega : \phi(x) > t\} = \Omega \setminus \bar{U}_t$ and $\{x \in \Omega : \phi(x) < t\}$ are open in Ω for each $t \in \mathbb{R}$ meaning ϕ is continuous on Ω . Since g is increasing and $g(R) = \alpha$ by equation (5.7), $\phi(x) \leq \alpha$ for all $x \in \Omega$ so $\phi \in M_\alpha$. Notice that this relies on the positivity of α .

The first equality follows from Lemma 5.3. By construction the level sets of ϕ , U_t , and the level sets of $\frac{|\nabla u^*|}{u^*}$, U_t^* , have equal measure so

$$\int_{U_t} \phi^2 dx = \int_{U_t^*} \left(\frac{|\nabla u^*|}{u^*} \right)^2 dx \quad (5.8)$$

for all $t \in (m, 1)$. The isoperimetric inequality asserts that the ball has minimal surface area among domains of equal volume so $|\partial U_t^*| \leq |\partial U_t|$ for all $t \in (m, 1]$. Since $\phi(x) = g(r(t)) \leq \alpha$ on Ω we have:

$$\begin{aligned} \int_{\partial U_t^*} \frac{|\nabla u^*|}{u^*} d\sigma &= g(r(t)) |\partial U_t^*| \\ &\leq g(r(t)) |\partial U_t| \\ &= g(r(t)) (|\Gamma_t| + |S_t|) \\ &= g(r(t)) |\Gamma_t| + \int_{S_t} g(r(t)) d\sigma \\ &\leq \alpha |\Gamma_t| + \int_{S_t} \phi d\sigma. \end{aligned} \quad (5.9)$$

Now compare the definitions:

$$H_\Omega[U_t, \phi] = \frac{1}{|U_t|} \left(\int_{S_t} \phi d\sigma + \alpha |\Gamma_t| - \int_{U_t} \phi^2 dx \right) \quad (5.10)$$

$$H_B \left[B_{r(t)}, \frac{|\nabla u^*|}{u^*} \right] = \frac{1}{|B_{r(t)}|} \left(\int_{\partial B_{r(t)}} \frac{|\nabla u^*|}{u^*} d\sigma - \int_{B_{r(t)}} \left(\frac{|\nabla u^*|}{u^*} \right)^2 dx \right). \quad (5.11)$$

The prefactors are equal, by equation (5.8) the final term in each line is equal, and inequality (5.9) finishes the proof. $\square$

Finally, we bring these results together to prove the Bossel-Daners inequality.

Proof of Theorem 5.2. This is just a matter of combining the ingredients. Assume that Ω is a bounded domain of class C^2 . If $\phi \in M_\alpha(\Omega)$ is the function we defined previously then by Theorem 5.5 there exists a $t \in (m, 1)$ such that $\lambda_1(\Omega) \geq H_\Omega[U_t, \phi]$. But then Lemma 5.10 implies

$$\lambda_1(\Omega) \geq H_\Omega[U_t, \phi] \geq H_B[B(r(t)), \phi^*] = \lambda_1^\alpha(B_R).$$

□

The proof can be extended to Lipschitz domains using an approximation of the domain by a sequence of C^∞ domains.

It was conjectured by Bareket that the optimising domain of the Faber-Krahn, Szego-Weinberger and Bossel-Daners inequalities, i.e., the ball, was the same for the $\alpha < 0$ regime. However, a counter-example from Freitas and Krejčířik in 2015 disproved this [11]. Recall that for $\alpha < 0$, $\lambda_1^\alpha(\Omega) < 0$ and so we seek maximising domains.

Theorem 5.11 (Freitas-Krejčířik, 2015). *Let $\alpha < 0$. Given a ball B_r of radius r , there exist r_1 and r_2 with $r_1 < r_2$ such that the spherical shell*

$$S_{r_1, r_2} := \{x \in \mathbb{R}^N : r_1 < |x| < r_2\}$$

with the same volume of B_r satisfies $\lambda_1^\alpha(B_r) < \lambda_1^\alpha(S_{r_1, r_2})$ for all $\alpha < 0$ sufficiently large in modulus.

The proof uses the asymptotic expansion of $\lambda_1^\alpha(\Omega)$ for $\alpha \rightarrow -\infty$ for Ω being a ball and Ω being a spherical shell. It then shows that for α sufficiently large, the latter domain has a larger eigenvalue. However it was also shown in the same paper that the Bareket conjecture holds in $\mathbb{R}^2$ for α sufficiently close to zero:

Theorem 5.12 (Freitas-Krejčířik, 2015). *Let $\Omega \subset \mathbb{R}^2$ be a bounded Lipschitz domain and Ω^* be the symmetric rearrangement. Then there exists $\alpha^* < 0$ depending on $|\Omega|$ such that*

$$\lambda_1^\alpha(\Omega) \leq \lambda_1^\alpha(B)$$

for all $\alpha \in [\alpha^, 0]$.*

In $\mathbb{R}^2$ there are maximisers for α sufficiently small and large in modulus which begs the questions: do these regimes seamlessly transition i.e., for all $\alpha < \alpha^*$ where α^* is minimal do we have that the annulus is the maximiser? The question remains open.

5.2 The Second Robin Eigenvalue

Now turning to the second eigenvalue, we look once again to the Dirichlet problem for inspiration and the following theorem due to Krahn [21] and Szegő [35].

Theorem 5.13 (Krahn-Szegő, 1926). *If $\Omega \subset \mathbb{R}^N$ is a domain and D is the disjoint union of two balls, each with volume $|\Omega|/2$, then*

$$\mu_2(\Omega) \geq \mu_2(D).$$

We provide the proof included in [18].

Proof. Let v be a Dirichlet eigenfunction with eigenvalue $\mu_2(\Omega)$. Denote the nodal domains of v by

$$\Omega^+ := \{x \in \Omega : v(x) > 0\},$$

$$\Omega^- := \{x \in \Omega : v(x) < 0\}.$$

By Corollary 4.9.3, both sets are non-empty and connected. Then considering the restriction of v to Ω^+ , v is an eigenfunction that doesn't change sign on Ω^+ so $\mu_2(\Omega) = \mu_1(\Omega^+)$ since the only

eigenfunction which doesn't change sign is the first. Then by the Faber-Krahn inequality (Theorem 5.1), $\mu_1(\Omega^+) \geq \mu_1(B^+)$ where B^+ is the ball with volume equal to Ω^+ . By the same reasoning, if B^- is the ball with volume equal to Ω^- then $\mu_2(\Omega) \geq \mu_1(B^+)$. Thus

$$\mu_2(\Omega) \geq \max\{\mu_1(B^+), \mu_1(B^-)\}.$$

The right-hand side is minimised when the eigenvalues are equal, that is when the balls have equal volume. Let B be the ball with volume $|\Omega|/2$ and $D = B \sqcup B$. Then

$$\mu_2(\Omega) \geq \mu_1(B) = \mu_1(D) = \mu_2(D).$$

□

This is a specific example of the broader Wolf-Keller theorem for Dirichlet eigenvalues which states that if an eigenvalue $\mu_n(\Omega)$ is minimised on a disconnected domain Ω then each of the connected components is a minimiser for a lower eigenvalue [43]. The same result holds for the Neumann eigenvalues [34] but it remains open for Robin eigenvalues. The following theorem of Kennedy fails to provide a counterexample, proved by Kennedy in his PhD thesis, whose doctoral supervisor was Daners. We follow Kennedy's proof [18].

Theorem 5.14 (Kennedy, 2009). *If $\Omega \subset \mathbb{R}^N$ is a domain and D is the disjoint union of two balls, each with volume $|\Omega|/2$, then*

$$\lambda_2^\alpha(\Omega) \geq \lambda_2^\alpha(D).$$

At first glance, it seems that we can recycle the proof for the Dirichlet case. However, the Bossel-Daners inequality (Theorem 5.2) applies only to Lipschitz domains and we do not know *a priori* whether the nodal domains are Lipschitz.

Assume Ω is a connected domain. Let u be an eigenfunction corresponding to the second eigenvalue $\lambda_2(\Omega)$. By Corollary 4.9.1, as Ω is connected, u changes sign inside Ω so that the sets

$$\begin{aligned}\Omega^+ &= \{x \in \Omega : u(x) > 0\}, \\ \Omega^- &= \{x \in \Omega : u(x) < 0\}.\end{aligned}$$

are non-empty. Let

$$u^+ = \max\{u, 0\}, \quad u^- = \min\{u, 0\}$$

be the non-negative and non-positive parts of u respectively. Let B^+ and B^- be the respective symmetric rearrangements of Ω^+ and Ω^- . Exactly as above our aim is to show that

$$\lambda_2^\alpha(\Omega) \geq \max\{\lambda_1^\alpha(B^+), \lambda_1^\alpha(B^-)\}.$$

Without loss of generality we will prove that $\lambda_2^\alpha(\Omega) \geq \lambda_1^\alpha(B^+)$. Problems arise where the nodal set meets the boundary as at such points the Robin condition implies $\frac{\partial u}{\partial \nu} = 0$. To rectify this we attach a thin strip to Ω^+ near $\partial\Omega$. Let $\varepsilon > 0$ and choose $\delta > 0$ such that the strip attached to the boundary,

$$S_\varepsilon := \{x \in \Omega : d(x, \partial\Omega) < \delta\}$$

has volume less than ε . Define the union of the strip and the nodal domain by $U_\varepsilon := \Omega^+ \cup S_\varepsilon$. Then ∂U_ε has an internal and external component: $\partial\Omega \subset \partial U_\varepsilon$ is the external part and define the internal boundary by $\Gamma_\varepsilon := \partial U_\varepsilon \setminus \partial\Omega$. Observe that every point on Γ_ε is at least a distance δ from $\partial\Omega$. A demonstration of the sets is given in Figure 10.

Now consider the eigenvalue problem on U_ε :

$$\begin{cases} -\nabla^2 u = \lambda^\alpha u & \text{in } U_\varepsilon, \\ \frac{\partial u}{\partial \nu} + \alpha u = 0 & \text{on } \partial\Omega, \\ u = 0 & \text{on } \Gamma_\varepsilon. \end{cases} \quad (5.12)$$

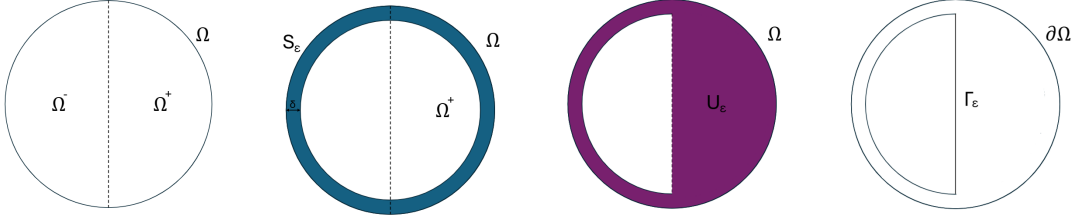


Figure 10: The sets Ω^+ , S_ε , U_ε and ∂U_ε .

We denote the space of weak solutions to (5.12) by H_{U_ε} which is given by the closure of $C_c^\infty(U_\varepsilon \cup \partial\Omega)$. Then the problem (5.12) has a smallest eigenvalue, say Λ_1^α , which is given by the Rayleigh quotient:

$$\Lambda_1^\alpha(U_\varepsilon) = \inf_{v \in H_{U_\varepsilon}} \frac{\int_{U_\varepsilon} |\nabla v|^2 dx + \alpha \int_{\partial\Omega} v^2 d\sigma}{\int_{U_\varepsilon} v^2 dx}. \quad (5.13)$$

Lemma 5.15. *For all $\varepsilon > 0$,*

$$\lambda_2^\alpha(\Omega) = \frac{\int_{U_\varepsilon} |\nabla u^+|^2 dx + \alpha \int_{\partial\Omega} (u^+)^2 d\sigma}{\int_{U_\varepsilon} (u^+)^2 dx}.$$

Proof. Multiplying the eigenvalue equation by an arbitrary $\phi \in H^1(\Omega)$ and integrating over Ω , we see that $\lambda_2^\alpha(\Omega)$ satisfies the weak formulation:

$$\int_{\Omega} \nabla u \cdot \nabla \phi dx + \alpha \int_{\partial\Omega} u \phi d\sigma = \lambda_2^\alpha(\Omega) \int_{\Omega} u \phi dx. \quad (5.14)$$

Letting $\phi = u^+ \in H_1(\Omega)$ be the trial function, we use that u^+ either agrees with u or vanishes so that

$$\nabla u^+ \cdot \nabla u = |\nabla u^+|^2, \quad uu^+ = (u^+)^2.$$

Since $\Omega^+ \neq \emptyset$, the L^2 norm of u^+ , is positive and so by rearranging equation (5.14),

$$\lambda_2^\alpha(\Omega) = \frac{\int_{\Omega} |\nabla u^+|^2 dx + \alpha \int_{\partial\Omega} (u^+)^2 d\sigma}{\int_{\Omega} (u^+)^2 dx}.$$

But as the integrands of the volume integrals are non-zero only on $\Omega^+ \subset U_\varepsilon$, we can restrict the domains from Ω to U_ε for any $\varepsilon > 0$. $\square$

Lemma 5.16. *For all $\varepsilon > 0$,*

$$\lambda_2^\alpha(\Omega) \geq \Lambda_1^\alpha(U_\varepsilon).$$

Proof. This follows from Lemma 5.15 and the variational characterisation in equation (5.13), provided $u^+ \in H_{U_\varepsilon}$. This is shown in [18] using the fact that H_{U_ε} is closed and expressing u^+ as the limit of functions in H_{U_ε} . $\square$

As Γ_ε may not be smooth, we approximate U_ε by a sequence of smooth domains $U_n \subset U_\varepsilon$ such that $\Lambda_1^\alpha(U_n) \rightarrow \Lambda_1^\alpha(U_\varepsilon)$.

Lemma 5.17. *There exists a sequence of Lipschitz domains $U_n \subset U_\varepsilon$ such that for all n ,*

1. $\partial\Omega \subset \partial U_n$,

$$2. \ d(\partial\Omega, \partial U_n \setminus \partial\Omega) > 0,$$

and as $n \rightarrow \infty$,

$$3. \ |U_\varepsilon \setminus U_n| \rightarrow 0,$$

$$4. \ \Lambda_1(U_n) \rightarrow \Lambda_1(U_\varepsilon).$$

Tools established, we are now prepared to prove Theorem 5.14. The proof will take the form of two limiting arguments: first, we approximate U_ε by a sequence of Lipschitz domains U_n so that we can use the Bossel-Daners inequality which is preserved in the limit $n \rightarrow \infty$. Then we combine this with the inequality from Lemma 5.16 and take the limit $\varepsilon \rightarrow 0$ to make a conclusion about the original domain.

Proof of Theorem 5.14. We can choose a sequence of Lipschitz domains U_n as per Lemma 5.17 and consider the Robin problem on U_n . As ∂U_n is Lipschitz, the Bossel-Daners inequality holds. Let B_n be the symmetric rearrangement of U_n . Then Theorem 5.2 states that

$$\lambda_1^\alpha(B_n) \leq \lambda_1^\alpha(U_n)$$

for all $n \in \mathbb{N}$. Since $H_{U_n} \subset H^1(U_n)$ we can compare the variational formulations of

$$\begin{aligned} \lambda_1^\alpha(U_n) &= \min_{v \in H^1(U_n)} \frac{\int_{U_n} |\nabla v|^2 dx + \alpha \int_{\partial\Omega} v^2 d\sigma}{\int_{U_n} v^2 dx}, \\ \Lambda_1^\alpha(U_n) &= \min_{v \in H_{U_n}} \frac{\int_{U_n} |\nabla v|^2 dx + \alpha \int_{\partial\Omega} v^2 d\sigma}{\int_{U_n} v^2 dx} \end{aligned}$$

and use the fact that the former minimum is taken over a larger set to conclude that

$$\Lambda_1^\alpha(U_n) \geq \lambda_1^\alpha(U_n) \geq \lambda_1^\alpha(B_n). \quad (5.15)$$

Let B_ε be the symmetric rearrangement of U_ε . Then using Lemma 5.17 part 3,

$$|U_n| \rightarrow |U_\varepsilon| \text{ as } n \rightarrow \infty \Rightarrow |B_n| \rightarrow |B_\varepsilon| \text{ as } n \rightarrow \infty.$$

As the first eigenvalue on the ball is a continuous function of its radius (as Bessel functions are continuous), $\lambda_1^\alpha(B_n) \rightarrow \lambda_1^\alpha(B_\varepsilon)$. The previous lemma also tells us that $\Lambda_1^\alpha(U_n) \rightarrow \Lambda_1^\alpha(U_\varepsilon)$. Therefore, taking the limit of inequality (5.15) as $n \rightarrow \infty$, we see that

$$\Lambda_1^\alpha(U_\varepsilon) \geq \lambda_1^\alpha(B_\varepsilon). \quad (5.16)$$

Finally, we shrink the strip S_ε by sending $\varepsilon \rightarrow 0$. Then $|U_\varepsilon| \rightarrow |\Omega^+|$ meaning that the symmetric rearrangements follow $|B_\varepsilon| \rightarrow |B^+|$. Again the continuity of eigenvalue with respect to radius means that $\lambda_1^\alpha(B_\varepsilon) \rightarrow \lambda_1^\alpha(B^+)$. Then Lemma 5.16 states that $\lambda_2^\alpha(\Omega) \geq \Lambda_1^\alpha(U_\varepsilon)$ which is bounded below by $\lambda_1^\alpha(B_\varepsilon)$ by equation (5.16). Taking the limit as $\varepsilon \rightarrow 0$ gives

$$\lambda_2^\alpha(\Omega) \geq \lambda_1^\alpha(B^+),$$

which proves the theorem as per our previous remark. $\square$

Lemma 5.18. *We can remove the assumption of connectedness in Theorem 5.14.*

Proof. Let Ω be a disconnected domain. Then either $\lambda_2^\alpha(\Omega) = \lambda_2^\alpha(\Omega_0)$ for some connected component Ω_0 of Ω or there exist disjoint connected components Ω_1, Ω_2 of Ω such that

$$\lambda_1^\alpha(\Omega_1) = \lambda_1^\alpha(\Omega) \leq \lambda_2^\alpha(\Omega) = \lambda_1^\alpha(\Omega_2).$$

In the first case, Ω_0 is a connected domain to which we can apply Theorem 5.14. Then

$$\lambda_2^\alpha(\Omega) = \lambda_2^\alpha(\Omega_0) \geq \lambda_2^\alpha(D') \geq \lambda_2^\alpha(D)$$

where D' is the disjoint union of two balls, each with half the volume of Ω_0 and D is the disjoint union of two balls, each with half the volume of Ω . The final inequality follows from the monotonicity with respect to homothetic scaling, Lemma 4.8. In the second case, $\lambda_2^\alpha(\Omega) = \lambda_1^\alpha(\Omega_2) \geq \lambda_1^\alpha(B_2)$ where B_2 is the symmetric rearrangement of Ω_2 by the Bossel-Daners inequality. Similarly $\lambda_2^\alpha(\Omega) \geq \lambda_1^\alpha(\Omega_1) \geq \lambda_1^\alpha(B_1)$. Therefore arguing as before,

$$\lambda_2^\alpha(\Omega) \geq \max\{\lambda_1^\alpha(B_1), \lambda_1^\alpha(B_2)\}$$

and the right-hand side is minimised when both eigenvalues are equal i.e., when B_1 and B_2 have equal volume and so

$$\lambda_2^\alpha(\Omega) \geq \lambda_1^\alpha(D) = \lambda_2^\alpha(D),$$

finishing the proof. $\square$

The minimising domain is disconnected so it might be reasonable to seek a minimising domain among connected domains, however no such minimiser exists. Let's connect up the disjoint balls in our domain $D = B_1 \cup B_2$ with a thin cylinder C_n with volume $1/n$ to create a connected 'dumbbell domain' Ω_n , shown in Figure 11. The cylinder C_n must be constructed from the balls so that $|\Omega_n| = |B_1 \cup B_2|$ and the boundary is still Lipschitz. Then as shown in [6], $\lambda_2^\alpha(\Omega_n) \rightarrow \lambda_2^\alpha(D)$ as $n \rightarrow \infty$, meaning that no connected minimiser can exist. For any candidate Ω^* , there would be some $n \in \mathbb{N}$ sufficiently large that $\lambda_2^\alpha(\Omega^*) > \lambda_2^\alpha(\Omega_n)$.

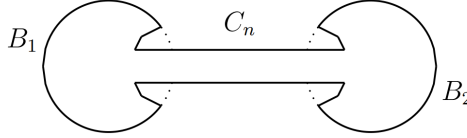


Figure 11: The domain Ω_n

6 Concluding Remarks

In Section 2.2 we proved that the cube extremises the first Robin eigenvalue among boxes of fixed volume as later on we would seek a minimiser when relaxing the box constraint. However due to the semi-explicit expression for the eigenvalues on a box, much more can be said and Laugesen collects many more results in [25] concerning the second Robin eigenvalue, such as convexity and monotonicity of $\lambda_2^\alpha(\Omega)$ and the spectral gap $\lambda_2^\alpha(\Omega) - \lambda_1^\alpha(\Omega)$. These results can be used to partially answer the question first posed by Mark Kac in 1966, ‘Can One Hear the Shape of a Drum?’ [17]. The question is an inverse spectral one: does the spectrum of the Laplacian uniquely determine its domain? In general $\mathbb{R}^n$ this question was already settled in 1964 when Milnor provided a pair of isospectral 16-dimensional tori [32]. It took until 1992 for a counterexample in $\mathbb{R}^2$ to be found. Gordon, Webb, and Wolpert constructed a pair of planar isospectral domains using a group theoretic argument by gluing together seven triangles in two different ways [14]. However these are examples of the Dirichlet and Neumann Laplacian and it was suggested by Arendt, ter Elst, and Kennedy in [1] that

‘it may well be the case that one can hear the shape of a drum after all, if one loosens the membrane before striking it.’

Laugesen proved that from the first two Robin eigenvalues on the rectangle in $\mathbb{R}^2$ we can recover the dimensions of the rectangle, giving a positive answer to Kac’s question. The Robin condition with $\alpha \neq 0$ is essential as no pre-specified number of Neumann eigenvalues can determine the dimensions of a rectangle [25]. The simplicity of the domain means that we only need the first two tones, whereas most results in this field require the entire spectrum. For example, it is possible to hear the area of a planar domain given the entire spectrum, thanks to Weyl’s law [38]:

Theorem 6.1 (Weyl’s Law, 1911). *For a domain $\Omega \subset \mathbb{R}^2$, the Dirichlet eigenvalues obey*

$$\lim_{n \rightarrow \infty} \frac{\mu_n}{n} = \frac{4\pi}{|\Omega|}.$$

McCartin has provided a thorough account of the spectrum of the Laplacian on the equilateral triangle and completely determined the domains on which eigenfunctions can be expressed as a finite series of trigonometric functions. These are domains which are constructed as the finite tilings of isosceles right-angled triangles and ‘semi-equilateral’ $(\pi/2, \pi/3, \pi/6)$ triangles [31].

The results for the extremisation of Robin eigenvalues on polygonal domains are incredibly recent (2019, 2023, 2024) and there are still many open questions concerning both the global extremisers and also whether the regular N -gon is the extremiser among N -gons of fixed area.

The variational characterisation of the eigenvalues in facts extends to general elliptic operators, as each elliptic operator has an associated bilinear form and so an associated Rayleigh quotient, $B[u, u]/(u, u)$, which is minimised by an eigenfunction. The basic conclusions we drew about the existence and positivity of the first eigenfunction and the simplicity of the first eigenvalue hold still for non-self-adjoint operators, in particular when the first eigenvalue is not necessarily real [8].

Regarding the monotonicity of the eigenvalues with respect to domain, Giorgi and Smits proved several positive results such as monotonicity with respect to a ball enclosing the domain [12].

Nodal geometry is a rich field and we touched only briefly on nodal domains. If an eigenfunction attains the upper bound of nodal domains then its eigenvalue is referred to as Courant sharp and Pleijel’s theorem tells us that for a fixed domain only finitely many Dirichlet eigenvalues are Courant sharp [26].

All of these are avenues I was interested to explore, if not limited by either the time constraints of the project or the limits which the Robin condition imposes on mathematical progress. Mathematics progresses through a process of generalisation, where new discoveries over time become the special cases and examples, and are replaced by more general overarching structures. While it is valuable to explore the special cases in depth, as these are where we can conclude the most, it is equally valuable to take inspiration from these examples to discover more about the general case.

I hope this approach has been evident, both in the progression from explicit symmetric domains to general ones, and in my treatment of the Dirichlet and Neumann problems being special cases of the more general the Robin problem.

References

- [1] Wolfgang Arendt, Antonius Frederik Maria ter Elst, and James Kennedy. “Analytical aspects of isospectral drums”. In: *Oper. Matrices* 8.1 (2014), pp. 255–277. ISSN: 1846-3886,1848-9974. DOI: 10.7153/oam-08-14. URL: <https://doi.org/10.7153/oam-08-14>.
- [2] Catherine Bandle. “Dido’s problem and its impact on modern mathematics”. In: *Notices Amer. Math. Soc.* 64.9 (2017), pp. 980–984. ISSN: 0002-9920,1088-9477. DOI: 10.1090/noti1576. URL: <https://doi.org/10.1090/noti1576>.
- [3] Marie-Hélène Bossel. “Membranes élastiquement liées: extension du théorème de Rayleigh-Faber-Krahn et de l’inégalité de Cheeger”. In: *Acad. Sci. Paris* (1986).
- [4] Almut Burchard. *A Short Course on Rearrangement Inequalities*. URL: <https://www.math.utoronto.ca/almut/rearrange.pdf>. (accessed: March 2025).
- [5] *CUHK Convex Optimisation Lecture Notes*. URL: https://www.math.cuhk.edu.hk/course_builder/2122/math4230/Note9.pdf. (accessed: March 2025).
- [6] Edward Dancer and Daniel Daners. “Domain perturbation for elliptic equations subject to Robin boundary conditions”. In: *J. Differential Equations* 138.1 (1997), pp. 86–132. ISSN: 0022-0396,1090-2732.
- [7] Daniel Daners. “A Faber-Krahn inequality for Robin problems in any space dimension”. In: *Math. Ann.* 335.4 (2006), pp. 767–785. ISSN: 0025-5831,1432-1807.
- [8] Lawrence Evans. *Partial differential equations*. Second. Vol. 19. Graduate Studies in Mathematics. American Mathematical Society, Providence, RI, 2010. ISBN: 978-0-8218-4974-3.
- [9] Lawrence Evans and Ronald Gariepy. *Measure theory and fine properties of functions*. Studies in Advanced Mathematics. CRC Press, Boca Raton, FL, 1992, pp. viii+268. ISBN: 0-8493-7157-0.
- [10] Georg Faber. “Beweis dass unter allen homogenen Membranen von gleicher Fläche und gleicher Spannung die kreisförmige den tiefsten Grundton gibt”. In: *Sitzungsbericht der bayrischen Akademie der Wissenschaften* (1923), pp. 169–172.
- [11] Pedro Freitas and David Krejčířík. “The first Robin eigenvalue with negative boundary parameter”. In: *Adv. Math.* 280 (2015), pp. 322–339. ISSN: 0001-8708,1090-2082.
- [12] Tiziana Giorgi and Robert Smits. “Monotonicity results for the principal eigenvalue of the generalized Robin problem”. In: *Illinois Journal of Mathematics* 49.4 (2005), pp. 1133–1143.
- [13] Carolyn Gordon and David Webb. “You Can’t Hear the Shape of a Drum”. In: *American Scientist, Volume 84* (1996).
- [14] Carolyn Gordon, David Webb, and Scott Wolpert. “One cannot hear the shape of a drum”. In: *Bull. Amer. Math. Soc. (N.S.)* 27.1 (1992), pp. 134–138.
- [15] Antoine Henrot, ed. *Shape optimization and spectral theory*. De Gruyter Open, Warsaw, 2017, p. 464. ISBN: 978-3-11-055088-7.
- [16] Antoine Henrot and Michel Pierre. *Shape variation and optimization*. Vol. 28. EMS Tracts in Mathematics. European Mathematical Society (EMS), Zürich, 2018. ISBN: 978-3-03719-178-1.
- [17] Mark Kac. “Can One Hear the Shape of a Drum?” In: *The American Mathematical Monthly, Vol. 73, No. 4, Part 2: Papers in Analysis* (1966).
- [18] James Kennedy. “An isoperimetric inequality for the second eigenvalue of the Laplacian with Robin boundary conditions”. In: *Proc. Amer. Math. Soc.* 137.2 (2009), pp. 627–633. ISSN: 0002-9939,1088-6826.
- [19] James Kennedy. “On the Isoperimetric Problem for the Laplacian with Robin and Wentzell Boundary Conditions”. PhD thesis. The University of Sydney, 2010.
- [20] Edgar Krahn. “Über eine von Rayleigh formulierte Minimaleigenschaft des Kreises”. In: *Math. Ann.* 94.1 (1925), pp. 97–100. ISSN: 0025-5831,1432-1807.

- [21] Edgar Krahn. “Über Minimaleigenschaften der Kugel in drei un mehr Dimensionen”. In: *Acta Comm. Univ. Dorpat.* (1926), pp. 1–44.
- [22] David Krejčířík, Vladimir Lotoreichik, and Tuyen Vu. “Reverse isoperimetric inequality for the lowest Robin eigenvalue of a triangle”. In: *Appl. Math. Optim.* 88.2 (2023), Paper No. 63, 33. ISSN: 0095-4616,1432-0606.
- [23] Gabriel Lamé. “Mémoire sur la propagation de la chaleur dans les polyédres”. In: *Journal de l’École Polytechnique* 22 (1833), pp. 194–251.
- [24] James Larsen-Scott and Julie Clutterbuck. “Local spectral optimisation for Robin problems with negative boundary parameter on quadrilaterals”. In: *J. Math. Phys.* 65.3 (2024), Paper No. 031505, 20. ISSN: 0022-2488,1089-7658.
- [25] Richard Laugesen. “The Robin Laplacian—Spectral conjectures, rectangular theorems”. In: *J. Math. Phys.* 60.12 (2019), pp. 121507, 31. ISSN: 0022-2488,1089-7658.
- [26] Michael Levitin, Dan Mangoubi, and Iosif Polterovich. *Topics in spectral geometry*. Vol. 237. Graduate Studies in Mathematics. American Mathematical Society, Providence, RI, 2023, pp. xviii+325. ISBN: 9781470475253.
- [27] Brian McCartin. “Eigenstructure of the equilateral triangle. I. The Dirichlet problem”. In: *SIAM Rev.* 45.2 (2003), pp. 267–287.
- [28] Brian McCartin. “Eigenstructure of the equilateral triangle. II. The Neumann problem”. In: *Math. Probl. Eng.* 8.6 (2002), pp. 517–539. ISSN: 1024-123X,1563-5147.
- [29] Brian McCartin. “Eigenstructure of the equilateral triangle. III. The Robin problem”. In: *Int. J. Math. Math. Sci.* 13-16 (2004), pp. 807–825. ISSN: 0161-1712,1687-0425.
- [30] Brian McCartin. “Eigenstructure of the equilateral triangle. IV. The absorbing boundary”. In: *Int. J. Pure Appl. Math.* 37.3 (2007), pp. 395–422. ISSN: 1311-8080,1314-3395.
- [31] Brian McCartin. *Laplacian eigenstructure of the equilateral triangle*. Hikari Ltd., Ruse, 2011. ISBN: 978-954-91999-6-3.
- [32] John Milnor. “Eigenvalues of the Laplace operator on certain manifolds”. In: *Proc. Nat. Acad. Sci. U.S.A.* 51 (1964), p. 542.
- [33] Dora Musielak. *Dido’s Problem. When a myth of ancient literature became a problem of variational calculus*. 2023. arXiv: 2301.02917 [math.HO]. URL: <https://arxiv.org/abs/2301.02917>.
- [34] Guillaume Poliquin and Guillaume Roy-Fortin. “Wolf-Keller theorem for Neumann eigenvalues”. In: *Ann. Sci. Math. Québec* 36.1 (2012), pp. 169–178. ISSN: 0707-9109.
- [35] George Pólya and Gabor Szegő. *Isoperimetric Inequalities in Mathematical Physics*. (AM-27). Princeton University Press, 1951. ISBN: 9780691079882.
- [36] John William Strutt Rayleigh Baron. *The Theory of Sound*. 2d ed. Dover Publications, New York, 1945, Vol. I, xlii+480 pp., vol. II, xii+504.
- [37] Walter Rudin. *Real and complex analysis*. Third. McGraw-Hill Book Co., New York, 1987, pp. xiv+416. ISBN: 0-07-054234-1.
- [38] Walter Strauss. *Partial differential equations*. Second. An introduction. John Wiley & Sons, Ltd., Chichester, 2008. ISBN: 978-0-470-05456-7.
- [39] Gabor Szegő. “Inequalities for certain eigenvalues of a membrane of given area”. In: *J. Rational Mech. Anal.* 3 (1954), pp. 343–356. ISSN: 1943-5282,1943-5290.
- [40] *UCLA Lecture Demonstration Manual*. URL: <http://www-cs-faculty.stanford.edu/~uno/abcde.html>. (accessed: March 2025).
- [41] George Watson. *A treatise on the theory of Bessel functions*. Cambridge Mathematical Library. Reprint of the second (1944) edition. Cambridge University Press, Cambridge, 1995, pp. viii+804. ISBN: 0-521-48391-3.

- [42] Hans Weinberger. “An isoperimetric inequality for the N -dimensional free membrane problem”. In: *J. Rational Mech. Anal.* 5 (1956), pp. 633–636. issn: 1943-5282,1943-5290.
- [43] Sven Wolf and Joseph Keller. “Range of the first two eigenvalues of the laplacian”. In: *Proc. R. Soc. Lond. A.* 447 (1994), pp. 397–412.

Appendix

A Python Code

Below is the code I wrote in Python to produce the rearrangement diagrams, Figure 7.

```
1 from matplotlib import pyplot as plt
2 import numpy as np
3
4 n = 1000 #Number of data points in domain
5 x = np.linspace(-4, 4, n) #Data points in domain
6
7 def f(x):
8     """The function we aim to rearrange
9     """
10    return (x**2 + 3*x + 2.5)*(x**2 - 1.5*x + 1)*np.exp(-0.5*x**2)
11
12 def levelset(f,t):
13     """find the level set of f, U_t.
14     """
15     lst = []
16     for x in np.linspace(-4,4,n):
17         if f(x)>t:
18             lst.append(x)
19     return lst
20
21 def maxx(f):
22     """finds the maximum of a function f over the domain
23     """
24     y = f(x)
25     z = list(y)
26     return max(y)
27
28 def endpoints(f,t):
29     """find the endpoints of the interval U_t^*
30     """
31     y = levelset(f,t)
32     if y != []:
33         eps = len(y)*8/n #the length of the domain is 8
34     else:
35         eps = 0
36     # if the level set is empty its rearrangement has length zero
37     return eps/2
38
39
40 y = f(x)
41
42 sdrx = [] #negative x-values in the symmetric decreasing rearrangement
43 sdry = [] #the corresponding y-values
44 sdrx2 = [] #positive x-values
45 sdry2 = [] #y-values for positive x-values
46
47 for t in np.linspace(0, maxx(f), 500):
48     sdry.append(t)
49     eps = endpoints(f,t)
50     sdrx.append(-eps)
51     sdrx2.append(eps)
52
53 #plot the original curve f
54 plt.plot(x,f(x),"r",label = "f(x)")
55 #plot the negative half of its symmetric decreasing rearrangement
56 plt.plot(sdrx,sdry,"k")
57 #plot the positive half
58 plt.plot(sdrx2,sdry,"k", label = "f*(x)")
59
60 t = 2.16 #the height at which we draw the level set
```

```

61
62 lst = levelset(f,t)
63 plt.scatter(np.array(lst),np.zeros(len(lst))+t,s=1)
64 #plot the original level set
65
66 eps = endpoints(f,t)
67 plt.plot([-eps,eps],[t,t],"g-") #plot the rearranged level set
68
69
70 plt.legend(loc="upper left")
71 plt.show()

```